SOP-24: Re-Ranking and Recovery Operations

Version 2.0 · 2026-08-08 · Owner: PPC Lead · Supersedes SOP-24 v1.0 (2026-07-24) in full · Extends SOP-23 and inherits its sizing chain by reference; everything here is the delta · Every threshold lives in the Master PPC Decision Criteria System v4.0; this SOP carries zero local thresholds · Verdict strings are quoted verbatim from the Keyword Decision Record Template v3.0 "Bands & Verdicts Reference" sheet, with the v4.0 Section 13.1 cross-map stated where the two diverge, per Ledger conflicts C10 and C18

1. Verdict and Scope

> VERDICT: a lost rank has a cause, and recovery spend released before the cause is resolved buys rank the cause takes back. Ranking and recovery run on identical thresholds and opposite disciplines: a ranking push starts from the question "can we win this position," a recovery starts from the question "why did we stop holding it," and a recovery that skips the second question is a ranking push wearing a cheaper velocity assumption. This SOP owns loss dating, the seven cause classes, the confound and void determination that SOP-06 and #35 both consume, baseline restoration and its verification artifacts, the observation window that separates a self-recovering rank from a bought one, recovery-adjusted sizing, the pre-OOS overlay and the post-restock standup, compressed in-flight verification, the recovery close, and the serial-recovery escalation out of PPC. It owns no threshold, no ladder rung, no push sizing chain and no campaign edit; each of those is named by number in Section 1.4.

1.1 What changed at v2.0

v1.0 was built to the superseded 10-element standard and measured 3 pages against a 10-18 page bar. v2.0 expands procedures from 5 to 14, failure modes from 5 to 13, and worked examples from 1 to 4; maps the full 25-verdict closed vocabulary rather than 5 rows; and adds Element 11 operator ramp and Element 12 interface contracts, neither of which exists in the 23 shipped SOPs. Section 10.2 carries the full change note. Four corrections of substance are made. First, v1.0's header cites "v4.0 Section 8.2" as the recovery authority; Section 8.2 is Syntax-Level Scenarios (S-Y) and carries no recovery content. Recovery is governed by Section 8.4 rule S-R13, Section 8.3 rule S-I5, and Section 11 rule T5, and every citation in this document is re-traced to those. Second, three v1.0 numerals attributed to v4.0 and not present there are retired rather than carried: the 2.0 positions-per-day recovery velocity reference, the day-4 checkpoint, and the third-recovery escalation count. Third, four v1.0 verdict strings belonging to no closed vocabulary are retired at Section 5.2. Fourth, the v1.0 citation of "#31 / P1.4" for out-of-stock forecasting is retired to the instrument layer per Ledger conflict C3, which names it as one of the five dangling citations that conflict records. All seven v1.0 cause classes and all five v1.0 account incidents carry forward, now sourced rather than asserted.

1.2 The seven cause classes, which are this document's primary asset

No other document in the library carries this list. SOP-06 v1.0 defers its confound cause classes to SOP-24 P1 by name, #35 defers its void classes to the same place, and the Ledger row for the diagnosis engine records SOP-24 P1 as one of the two entry points into it. The list is closed at seven; a new confirmed class increments this SOP under Section 10 and propagates to the three consumers named above.

Class	What dates it, and the evidence that names it	Consequence for the recovery, and who resolves the cause
OOS event	T5 Product Log event type "OOS start-end per variation" against the T2 **OOS Overlay Flag** and the daily rank series. The rank series inflects within the stockout window rather than before it.	The only class that resolves by an arrival rather than by a decision. Restoration is the restock itself, so P4 verification is the landed-stock artifact and P11 deploys at day 0. v4.0 Section 3 family F3 states the recovery contract directly: a term that held rank 6 before a stockout is a RECOVERY candidate with a proven ceiling under rule D10, not a never-ranked hopeful.
Price move	T5 event type "price move" on our side, or the Price Position Index (E10) crossing a class boundary on the market side. v4.0 Section 3 family F5 bands PPI as ADVANTAGED under 0.92, PAR 0.92-1.08, PREMIUM 1.08-1.25, OUTPRICED above 1.25.	v4.0 Section 7 rule D6 puts the price check first in every conversion diagnosis, so this class is tested before the four below it regardless of which looks likelier. OUTPRICED routes to pricing and vetoes listing spend as the primary remedy. Restoration is the price returning to PAR or better; PPC restores nothing. Live constraint: T7 field group D3 SQP median price reads MISSING, so the market limb of this test is not computable. WE-2 works the consequence.

Class	What dates it, and the evidence that names it	Consequence for the recovery, and who resolves the cause
Review shock	T2 rating and review count against the Review Position Gap (E11), which v4.0 Section 3 family F4 sets at FAR-BEHIND on under 50% of the top-5 median review count or 0.3 stars or more below it.	FAR-BEHIND fails the O3 competitor-credibility clause outright, so the recovery cannot be funded at any bid while it holds. The gap closes on review velocity, not on spend. Routes to the brand-management owner. BEHIND permits recovery on long-tail and mid-volume terms only while velocity closes it.
Listing event	T5 event types "listing / image / A+ change" and "eligibility flip". v4.0 family F4 flags CVR and CTR reads PROVISIONAL for any main-image or price-adjacent change inside 14 days.	Restoration is the edit reverted or the suppression lifted, verified with the artifact. Gate G5 governs the indexing limb: a term that fell out of index ranks at no bid level, and a rank-objective term moved to backend-only is a listing task before it is a bidding task. Routes to the listing owner.
Competitor surge	T2 **Competitor Depth (D8)** block: a competitor deal window, a launch, a price cut or a review surge on a top-10 holder. v4.0 Section 7 rule D7 splits market inflation from self-inflicted inflation through E1 before any aggressiveness verdict.	The one class with no restoration step, because nothing of ours changed. The decision is whether the position is still winnable on the O3 competitor-credibility test at the new market state, and a recovery funded into a competitor deal window buys the worst comparison of the month under S-C5. Live constraint: D8 reads MISSING, so this class is evidenced only where T5 carries a manually logged competitor entry.
Deal hangover	T5 event type "deal window" with the trend-exclusion flag set, read against the #32 settle marks. Gate G10 excludes event weeks and the 14 days following from trend verdicts.	Not a recovery case. Per the Ledger row for #32, a post-deal dip inside the settle is a hangover resolving by calendar rather than a loss to be recovered, and v4.0 family F13 classes it TRANSIENT at 1-2 weeks. This SOP holds and observes; it opens only if the depression survives settle expiry. Funding a hangover pays for motion the calendar was going to return anyway.
Taper too deep	The T6 decision log: the rank series inflects on a dated ladder step. Cross-read the T4 **Ladder State (step / date)** and **Discovered Floor $** against the T2 current bid.	The only class where PPC is the cause, and the only one whose remedy is a rung rather than a push. v4.0 Section 10 rule M3 states it: on slippage beyond tolerance, restore one step, and that value is the recorded minimum viable CPC. WE-4 computes the cost of missing this class at 59.4 times the correct remedy. Also read the #37 rule change log here: a runaway automation rule stepping against a stale target produces the same signature without a human ladder entry.

UNKNOWN is the eighth outcome and it is a legal finding. Carried forward from v1.0 unchanged: a cause that cannot be named with evidence routes to a deeper read, never to a default push. The classes are tested in the order above, which is not the order of likelihood but the order v4.0 requires: rule D5 puts the attribution check against the product log ahead of any lever explanation, and rule D6 puts the price check ahead of every other conversion cause.

1.3 Why the diagnosis is the document

SOP-23 exists because pushes were unsized. This document exists because losses were undiagnosed, and the two failures cost differently. An unsized push leaks at a rate; an undiagnosed loss buys a rank that reverts, so the entire recovery budget returns nothing and the term re-enters the queue at the next cycle looking like a fresh candidate. WE-4 puts the arithmetic on it: a rank slip caused by our own ladder stepping past its floor costs $6.60 per week to correct as a one-rung restore under rule M3, and $392.00 per week to correct as a sized recovery push, a factor of 59.4 and $20,040.80 a year on a single term if the class is read wrong every time. That ratio is the whole argument for P1 running before P7 and for P7 refusing to run without it. v4.0 Section 4 rule X9 states the principle in one line, and this SOP is that line made executable: the product log is consulted before PPC is credited or blamed for any swing.

1.4 Boundaries, with the owning component named for every excluded decision

Per the Cross-Reference Ledger, this component owns baseline restoration, the pre-OOS overlays, and the void and confound cause classes. Everything below is referenced by number and one clause and is never restated here.

Excluded decision	Owner	The seam, stated so it is owned
Every threshold, band, scenario and verdict issuance	v4.0 (#8)	A numeral in this SOP without a v4.0 section beside it is a defect. Section 10.3 lists the numerals v4.0 does not yet carry; they are amendment items, not local values.
Push candidacy, the O3 gauntlet, the M2 sizing worksheet, funding, in-flight operation and the achievement and stability confirmation	SOP-23	This SOP extends SOP-23 and inherits its worksheet chain by reference rather than restating it. The seam is dated: a rank that was held and then lost belongs to this SOP from the moment the loss is dated at P1, and returns to SOP-23's operating cadence once the recovery worksheet is funded. SOP-23 classifies the term at its candidacy gauntlet and hands over; this SOP never re-runs that classification, it consumes it.
Ladder rungs, step derivation, exit-bid discovery, the reversion rule, the floor register and modifier sizing	#28	#28 is the authoritative ladder that SOPs 12, 23, 24, 25 and 26 all reference. This SOP decides whether a taper happens and when, at P9 and P13; every rung value, every floor and the reversion arithmetic are #28's. A step size written into this document is a defect.
Campaign creation, bid and budget edits, the re-point execution and taper execution at campaign level	SOP-12	This SOP issues the standup order and the restore instruction carrying verdict-borne values; SOP-12 executes. A bid value invented by this SOP is a defect.
The post-objective normalization sequence and the descent schedule	SOP-25	A recovery that reaches its baseline exits into SOP-25 exactly as an achieved push does. The seam runs the other way too: per the Ledger row for #25, its failure mode F2 exists precisely because a descent slip and a recovery get confused, and this SOP's cause classes are what separate them.
Floor operations, the quarterly re-ladder, the code-red protocol and Recovery Mode posture	SOP-26	Product-level Recovery Mode and keyword-level recovery are different objects that share a word. This SOP reads the product posture as a constraint at P8 and never declares, runs or closes the protocol.
Event plans, caps, arming, the settle marks and the hangover class	#32	A post-deal dip inside the 14-day settle is #32's hangover, not a loss. This SOP opens only at settle expiry, and cites the Ledger row rather than a #32 procedure number.
Family term ownership and the lead declaration	#33	Where a recovering term is contested, gate G11 holds the restore step until the lead is assigned. This SOP supplies the recovery class and cost-to-close for the family ordering and adjudicates neither side.
Out-of-stock forecasting, restock dates and cover	Instrument layer	v1.0 cited "#31 / P1.4" here. Per Ledger conflict C3, #31 carries no procedural document, so this SOP names the instrument and the feed: T2 **Inventory Band**, **Days to Stockout** and **Targeted Variation(s)**, fed daily from the Economics and Context Pack inventory feed. The dangling citation is retired and logged at Section 10.3.

Excluded decision	Owner	The seam, stated so it is owned
Queue ranking, envelope law, budget release and return	Instrument layer	T0 zone H and Account Rollup tab A2. This SOP submits a sized worksheet and receives a dated release; it never ranks the queue and never cites a #31 procedure number.
Negation walls at every structure event	#29	A restored or re-pointed campaign needs its walls rebuilt in the same working session. This SOP names the requirement and never the wall level.
Staging, upload and the T+1 verification pass	#38	Every change this SOP orders is staged and proven by the next-day diff. No console edit outside the staged set.
Prediction scoring rubric and tolerances	#6	This SOP supplies the confound cause classes that make a VOID legal and the diagnosis score; #6 owns the rubric that consumes them.
Which units are terminal or LTSF-tiered	#27	Per Ledger conflict C6 the deciding document sits outside this corpus. A CONSTRAINED earning-potential class bars recovery pushes outright under v4.0 Section 8.1 rule S-A10, and this SOP stops at that boundary.

2. Trigger Conditions

Each trigger names the artifact that detects it and the maximum allowed lag from signal to procedure start. Where v4.0 states the lag, the section is cited. Where it does not, the cell says so and the item appears at Section 10.3; no lag is invented here.

Trigger	Detecting artifact and the observable	Maximum lag to procedure start	Proc
Rank loss on a held term	T2 **Organic Rank** moving from AT or NEAR to a worse position class against **Best Rank (Hist) + Date**, read on the daily rank series (D9).	Not stated in v4.0. Family F6 bands position classes and velocity but sets no loss threshold and no detection interval. Amendment item AM-1, the highest-priority item in this register because it is the trigger of the entire document.	P1
Rank slipping at a held position	T2 **Velocity (pos/wk)** reading SLIPPING, which family F6 sets outside the FLAT band of plus or minus 1 position per week.	One weekly cycle. v4.0 Section 14 rule V4 sets the weekly full audit as the cadence producing Master Action Records.	P1
Inventory RED forecast	T2 **Inventory Band** flipping to RED, or **Days to Stockout** falling to 14 or fewer on a ranked term's targeted variation.	Same day. v4.0 Section 8.3 rule S-I3 and family F3 both set the pre-OOS taper trigger at 14 projected days.	P9
Stockout occurred	T2 **OOS Overlay Flag** set with the variation unbuyable, despite the P9 taper.	Same day. Rule S-I4 moves push and share objectives to Paused-pending and marks the trend windows for exclusion.	P10
Restock confirmed	The out-of-stock variation returns to in-stock on the inventory feed, cross-read against T2 **Targeted Variation(s)**.	Same day. Family F3 states the recovery play triggers at restock confirmation; rule S-I5 makes the play conditional on a pre-OOS baseline existing.	P11
Cause resolution	The P4 restoration artifact lands: stock in, price at PAR or better, listing edit reverted, suppression lifted, or the ladder rung restored.	Not stated in v4.0 for the interval between artifact and observation start. Amendment item AM-3.	P5
Observation window closes	T2 **Organic Rank** still depressed against the baseline after the P5 window, with the restoration artifact on file.	Not stated in v4.0 as a per-cause-class window. Family F13 carries persistence classes at syntax grain only. Amendment item AM-2.	P6

Trigger	Detecting artifact and the observable	Maximum lag to procedure start	Proc
Recovery funded	T0 zone H entry state changing to FUNDED with a dated release.	Same day for the post-restock case, where family F3 ties the play to the restock confirmation. For every other cause class the lag is not stated; amendment item AM-8.	P11
Daily in-flight verification	The daily rank series on a live recovery, read against the worksheet's velocity line and the T2 **In-Budget %**.	Same day. v4.0 Section 11 rule T5 sets daily rank checks for 3 weeks as the recovery mode's own cadence, and rule V4 places oversight-class checks on the daily cadence during pushes.	P12
ETA checkpoint	T2 **ETA Checkpoint** date arriving with the baseline not reached.	Same weekly read. Rules S-R13 and S-I5 both set the recovery ETA checkpoint at 3 weeks.	P13
Baseline reached	T2 **Organic Rank** at or better than the pre-OOS baseline recorded in T2 **Pre-OOS Baseline**.	Same weekly read. Rule T5 sets baseline reached as the recovery mode's exit; rule S-R6 separately sets 2 consecutive weeks at or under target rank as the taper trigger. Where baseline and target differ these resolve at different ranks; see AM-4.	P13
Confound detected	T5 trend-exclusion flag, an armed window from #32, or a dated competitor event landing inside a live recovery's read window.	Same weekly read. Gate G10 excludes event weeks and the 14 days following from trend verdicts; rule S-E5 makes rank achieved during events provisional.	P3
Repeat loss on one term	T6 carrying a prior closed recovery on the same term inside the review period.	Not stated in v4.0. No recovery-count rule and no window exist. Amendment item AM-6.	P14
Product posture flip	T0 zone A entering push freeze or code red (rules S-A3, S-A4).	Same cycle on S-A3. S-A4 enters Recovery Mode, whose cadence is daily under rule R4 and which permits new Ranking objectives only on RECOVERY-class terms at HIGH confidence under rule R3.	P8

3. Preconditions and Gates

Nothing in Section 4 runs until the manifest rows below read current on T7. Each row names the field group by its Catalog ID, the gate it feeds by its v4.0 number, and the observable that proves the gate passed. Where the operating corpus uses a different gate number for the same idea, the corpus alias is given at first use per Ledger conflict C9; the v4.0 number governs and the two schemes are never mixed inside one reason string.

Live manifest state, and what it costs this component. The T7 Manifest in the current Command Workbook template reads a packet completeness score of 0.571 , below the 85% DEGRADED floor that v4.0 Section 5 gate G8 sets. Under G8 a verdict whose gate fields are missing downgrades to a flag and the analysis states what it could not see. Five field groups this SOP depends on are not fully present, and two of them sit on the diagnosis rather than on the sizing, which is a harder failure: D3 SQP median price is MISSING , which removes the market limb of the price cause class that rule D6 requires first; D8 competitor depth is MISSING , which removes the evidence for the competitor-surge class; D9 daily rank series is PARTIAL on head terms only , which is the instrument that dates the loss at P1 and the instrument rule T5 requires daily for 3 weeks; D2 keyword-variation map is PARTIAL , which T7 itself records as blocking REPOINT VARIATION verdicts; and D1 per-variation margin is PARTIAL , carrying the G9 PROVISIONAL tag onto every ceiling in this document. P2 step 8 and P7 step 8 enforce the statement of these gaps on the record itself.

Manifest ID	Gate	Requirement before the dependent procedure runs	The observable that proves it passed
A4, A5	G1	No impossible-metric flag on the row: no CVR of zero with orders present, no ACOS without orders, no cross-table disagreement.	T2 **Flag: Zero-Quarantine** clear and **Validation State** not QUARANTINE. A quarantined row produces a flag and a data-refresh task, never a recovery. Rule P8 makes this beat urgency: a stockout in progress does not license a diagnosis on broken data.
A4, A1	G2	Sample sufficiency for every rate the recovery worksheet consumes: 15 or more clicks in 7 days for a bid read, 100 or more clicks for the CVR that sizes the recovery.	T2 **Clicks 7/30/60d** and **Sample State**. The recovery case fails this gate by construction more often than any other: a term that went out of stock or fell out of index has no recent clicks. Below 100 the CVR input falls back to the 60-day window, and where the outage consumed that window too, the pre-OOS baseline CVR is the stated input under test rather than a validated read. A1 reads MISSING, so four-period trend context is unavailable.
A3	G3	The hosting campaign is not budget-truncated. In-budget below 70% of the day marks every rate on that campaign TRUNCATED and forces the budget lever before any bid conclusion.	T2 **In-Budget %** at or above 70. Corpus alias: the KDR Bands sheet labels 51-70 "strained"; G3's 70% line governs. A3 reads PARTIAL, so the day-of-week curve behind G7 is unavailable.
A7	G4	The advertised offer is eligible and unsuppressed. An eligibility flip is itself a cause class, so this gate is read as evidence at P2 before it is read as a precondition at P11.	T2 **Eligibility** reads eligible. A flip halts all execution for the target the same day and opens the listing task.
B1	G5	The term is indexed and, on a rank objective, placed in title or bullets. No bid level ranks an unindexed term.	T2 **Indexed?** reads yes and **Listing Placement** reads title or bullets. Read as a cause first: a term that fell out of index has a listing cause, and re-indexing is the restoration artifact rather than a precondition someone waits out.
D1, D2	G6	The targeted variation reads GREEN before a recovery deploys. GREEN requires days on hand above lead time plus 21 days at max-sales-day velocity.	T2 **Inventory Band** and **Days to Stockout** read against **Targeted Variation(s)**. YELLOW freezes recovery spend at current level under rule S-I2; RED routes to P9 rather than to a recovery. D2 reads PARTIAL, so the declared backup chain is confirmed on head terms only.
A3	G7	Campaign budgets reconcile to the product envelope and product pacing sits within 25% of the day-of-week curve.	T2 **Envelope Status** and T0 zone H available dollars. A recovery that does not fit the envelope is a queue decision at the instrument layer, not a smaller recovery here.
All	G8	The packet manifest reports present, stale or missing before analysis. Below 85% of gate-relevant fields the packet is DEGRADED and produces flags only.	T7 **Status** column and the completeness rollup stamped on the cycle header. The live read is 0.571; see the note above for the five groups this component loses.

Manifest ID	Gate	Requirement before the dependent procedure runs	The observable that proves it passed
D1	G9	Price stability: no price change on the targeted variation inside the trailing 7 days, or the CVR and ACOS reads carry PROVISIONAL and no conversion verdict issues on them.	T5 **Event Type** filtered to price move, cross-read against T2 **Margin $/ Unit**. Corpus alias: SOP-12 v2.0, the Data Architecture and several operating SOPs use G9 for margin freshness; v4.0 assigns G9 to price stability and both states are checked here, because a price move is simultaneously a gate failure and a cause class.
D7	G10	Event weeks and the 14 days following are excluded from the trend reads that date a loss or score a recovery.	T5 **Trend-Exclusion Flag** and T2 **Event Mode**. Corpus alias: the operating corpus uses G12 for the event freeze and G10 for objective tagging; v4.0 assigns G10 to event exclusion. This gate is the void determination's instrument and P3 runs on it.
D6	G11	No second sibling carries an active push or recovery objective on the same term. Two siblings restoring one term is a portfolio decision.	T2 **Family Conflict** and **Term Owner (Family)**, sourced from Account Rollup tab A1. D6 reads PARTIAL, and the Bamboo root ownership was left OPEN at cycle 1, which is a live instance rather than a hypothetical.
B11	G12	The term exists in the current Master Keyword List with a syntax tag and a relevancy score, and the list is not stale past 30 days.	T2 **Syntax Tags** and **Relevancy** populated. Corpus alias: G12 is used for the event freeze across the operating corpus; v4.0 assigns G12 to keyword-list hygiene.
B3, D9	Rank pack	Best rank with its date, the pre-OOS baseline, velocity and ETA present for the term, with a daily rank series available for dating and in-flight verification. Not a numbered gate; a precondition without which P1 cannot run at all.	T2 **Best Rank (Hist) + Date**, **Pre-OOS Baseline**, **Velocity (pos/ wk)**, **ETA to Target (wks)**. Both read PARTIAL on head terms only. A term outside head-term coverage cannot have its loss dated and does not proceed past P1; the finding attaches to the T7 D9 row rather than resolving into a weekly-resolution guess.
B8, D1	Economics chain	T0 zone C live: AOV, contribution per order, break-even ACOS, Target ACOS, Max Acceptable and the Expected CPC reference. Not a numbered gate; a precondition without which no ceiling in Section 4 computes.	T0 zone C margin table version and its fresh or provisional state. B8 reads PRESENT on blended margin; D1 reads PARTIAL, so every ceiling here carries PROVISIONAL until the per-variation table lands. This is external dependency D1 in the Ledger, open, and it caps this component by name.
D3, D8	Cause evidence	SQP median market price and competitor depth present for the term, without which two of the seven cause classes cannot be evidenced.	T2 **Price Competitiveness Flag** and **Competitor Depth (D8)** carrying values rather than build markers. Both read MISSING account-wide. While they do, P2 can return UNKNOWN on a term whose cause is in fact readable from data the account does not yet hold, and P2 step 7 requires that state to be written as a manifest finding rather than as a diagnostic conclusion.

4. Step-Level Procedures

Fourteen procedures cover the recovery lifecycle from the moment a loss is dated to the moment its post-mortem is filed. Every step names the tab, the column and the value it reads or writes. A person who has never run a recovery executes these from the page. Time estimates assume the packet has refreshed and the OVERSIGHT filter is applied. P1 to P8 are the diagnosis and sizing spine and run on every cause class. P9 to P11 are the out-of-stock lane. P12 to P14 close every recovery without exception.

P1. Loss dating (mandatory first step, and it precedes every other read)

Per loss, 12-20 minutes. Source: v4.0 Section 4 rule X9 and Section 7 rule D5. Nothing in P2 is legal until the loss carries a date.

1. Open T2 and confirm T0 and T1 have been read first. Product posture and syntax caps constrain everything below them, and a recovery sized under an unread S-A3 push freeze is void at P8 regardless of its arithmetic.
2. Read T2 **Best Rank (Hist) + Date** and **Pre-OOS Baseline** and write both into the record before reading anything current. The question this procedure answers is not "how bad is rank now" but "when did it stop being what it was", and starting from the current number is how a recovery gets mistaken for a cold start.
3. Pull the daily rank series (D9) across the trailing 12 months that rule D10 reads, and locate the inflection: the first date on which the series leaves its held band and does not return. Write the inflection date to T2 **Situation** . Where D9 covers the term at daily resolution the date is exact; where it does not, the series is weekly and the date carries a plus or minus 7 day interval that travels with every downstream comparison in this procedure . Record which of the two applies. Do not average a weekly series into a false daily date.
4. Establish the loss magnitude in position classes rather than in positions: family F6 sets AT at or under target, NEAR from target to target plus 10, MID at 11-30 absolute, FAR above 30, and UNRANKED. A term moving NEAR to MID and a term moving AT to UNRANKED are different decisions, and the class transition is what the record states.
5. Read T2 **Velocity (pos/wk)** and the 4-week trend. Family F6 sets SLIPPING outside the FLAT band of plus or minus 1 position per week. A term still slipping has a live cause; a term that fell and stabilised has a resolved or one-time cause, and the two take different observation windows at P5.
6. Open T5 Product Log and filter to the inflection date plus or minus the dating interval from step 3. List every event in the window across all eight closed event types, with none excluded for looking unrelated. Rule D5 is explicit that the log is consulted before PPC is credited or blamed, and the log is read for the window rather than searched for a confirming entry.
7. Where the window is empty, say so on the record. An empty log window is evidence, not an absence of evidence: it removes four of the seven cause classes at a stroke and points the diagnosis at competitor surge or taper too deep, neither of which produces a T5 entry.
8. Write the attribution statement before proceeding. One line naming the inflection date, its interval, the position-class transition, the events in the window and the classes those events do and do not admit. PPC is neither credited nor blamed until this line exists, which is the point of the step and the reason it is first.

P2. Cause classification against the seven classes

Per loss, 20-30 minutes. Source: Section 1.2, tested in the order v4.0 rules D5 and D6 require. Exactly one primary cause is named, or UNKNOWN.

1. Price first, always. Rule D6 puts the Price Position Index ahead of every other conversion cause, because money is not spent fixing images when the market is paying materially less for the same promise. Read T2 **Price Competitiveness Flag** for our side and the SQP median market price for the market side. Where D3 reads MISSING the market limb is not computable; compute the limb you can, name the limb you cannot, and carry the gap into step 7.
2. OOS. Compare the inflection date to the T5 stockout window on the targeted variation. The test is directional and it is the one operators get backwards: an inflection _inside_ the window admits the class, an inflection _before_ the window rules it out, and a stockout that began after the rank fell is a coincidence the log happens to record. WE-2 works an instance where four days decide it.
3. Listing event. Read T5 for listing, image, A+ and eligibility events in the window, and re-read gate G5 as evidence: **Indexed?** reading no, or **Listing Placement** reading backend, is a cause and not merely a blocked precondition. Family F4 flags CVR and CTR PROVISIONAL for 14 days after any mainimage or price-adjacent change, so a conversion read inside that window is not admissible as counter-evidence.
4. Review shock. Read the Review Position Gap. FAR-BEHIND at under 50% of the top-5 median review count or 0.3 stars or more below it fails the O3 competitor-credibility clause outright, which means the class is not merely a cause but a bar on funding for as long as it holds.
5. Competitor surge. Read T2 **Competitor Depth (D8)** for a deal window, a launch, a price cut or a review surge on a top-10 holder, and run the E1 decomposition that rule D7 requires before any aggressiveness verdict: market-driven movement and self-inflicted movement take opposite responses. Where D8 reads MISSING, this class is evidenced only from a manually logged T5 competitor entry, and its absence is a coverage gap rather than a negative finding.
6. Deal hangover, then taper too deep. Read T5 for a deal window with its trend-exclusion flag: a dip inside the settle is #32's hangover and this procedure stops, records TRANSIENT per family F13, and hands to P5 for observation only. Then read T6 for a dated ladder step at the inflection, and the #37 rule change log for an automated step against a stale target. A rank series that inflects on our own ladder date is the taper-too-deep class and its remedy is a rung, not a push.
7. Name one primary cause with its evidence, or name UNKNOWN. Where two classes both fit, the one with the dated artifact wins and the other is recorded as a contributing condition; a diagnosis naming two primaries scores nothing at P14 because neither can be falsified. Where no class carries evidence, write UNKNOWN and route to the deeper read at P3 rather than to a default push. Where a class could not be tested because its feed is missing, the record states "not testable" rather than "not found", and the two are never collapsed, because the second is a diagnostic conclusion and the first is a manifest finding.

8. Write into T2 **Situation** every gap the classification rode over: an unevidenced price limb under D3, an unevidenced competitor limb under D8, a weeklyresolution inflection date under D9, a provisional margin under D1. Under gate G8 the analysis states what it could not see before it recommends anything, and hiding a fallback is a process violation under v4.0 Section 13.3 rule J7.

P3. The confound and void determination

Per loss and again per in-flight read, 8-12 minutes. Source: gate G10 and Section 8.12 rule S-E5. This is the procedure SOP-06 and #35 consume by name.

1. Test the dating window and the read window separately. A confound that sits on the inflection date corrupts the diagnosis; a confound that sits on a checkpoint corrupts the score. The same event can do one, both or neither, and the record says which.
2. Apply gate G10: event weeks and the 14 days following are excluded from trend verdicts. Read T5 **Trend-Exclusion Flag** and T2 **Event Mode** across both windows and compute the guard end date explicitly rather than by eye.
3. Where a confound falls inside a read window, the read is VOID and not a MISS. The distinction is the whole reason this procedure exists: a VOID protects the decider from luck in both directions, and misfiling a confounded read as a MISS teaches the calibration series that a sound assumption was wrong.
4. Classify the confound by its class so the consumers can route it: event or deal window (gate G10), competitor deal window (rule S-C5), price test in progress (gate G9), listing change inside 14 days (family F4), or a stockout on an adjacent variation shifting the traffic mix (rule S-I6). These five are the classes this SOP publishes.
5. Where rank is reached inside an armed window, rule S-E5 governs and the achievement is provisional: the two-week at-target clock under rule T1 starts only after the exclusion window closes, and no taper begins on the event spike.
6. Hand the void and its class references to #6 for the score and to #35 where an incrementality read was riding the same window. A confounded read handed over as clean is worse than no read, because the consumer cannot detect the contamination and files a conclusion against it.
7. Where a confound voids a read, re-date the checkpoint to the guard end rather than deeming the recovery failed. v4.0 states no re-dating rule and no ceiling extension for the suppressed interval; the extension is recorded as an input under test and logged at AM-5. WE-3 computes what one instance costs.

P4. Baseline restoration and its verification artifact

Per resolved cause, 10-25 minutes depending on class. No recovery spend deploys before this procedure closes.

1. Restore what changed, by class: stock landed, price returned to PAR or better on the Price Position Index, listing edit reverted or re-approved, suppression lifted, index confirmed, or the ladder rung restored one step per rule M3 with the rung value supplied by #28.
2. Verify with an artifact, never with an assertion. The artifact is named per class and filed against the record: the inventory feed showing the variation in stock and buyable, the T5 price-move entry showing the restored value with its PPI recomputed, the listing owner's confirmation with the live page state, the eligibility flag reading eligible, or the T6 row carrying the restored rung with its prior and new values. A recovery standup with no verification artifact is the F1 condition.
3. Where the cause class is competitor surge, there is nothing of ours to restore. Record the class as resolved-by-nothing and go directly to the O3 credibility retest at P6: the question is whether the position is still winnable against the new market state, and a recovery funded against a competitor whose price cut is permanent is a subscription rather than a recovery.
4. Where the cause class is deal hangover, restoration is the calendar. Record the settle expiry date and hold; do not restore, do not size, and do not open a recovery before that date.
5. Re-run gate G9. A price restoration is itself a price change inside the trailing 7 days, so the CVR and ACOS reads that would validate the restoration carry PROVISIONAL for that period and no conversion verdict issues on them. This is not a reason to wait; it is a reason the P5 observation window reads rank rather than rate.
6. Where the restoration was a listing or indexing fix, re-verify gate G5 rather than assuming the fix propagated. v4.0 states no re-check interval after an indexing fix and the platform does not confirm one; the interval used is recorded as an input under test and logged at AM-9.
7. Write the restoration record to T5 as a dated event and to T6 with its verification artifact referenced. The next cycle's P1 reads this entry, so a restoration that is not logged becomes an undated cause the following quarter.

P5. The observation window and the self-recovery test

Per restored cause, 5 minutes to open and 10 minutes to read. The cheapest procedure in the document and the one most often skipped.

1. Open the window on the restoration date and read rank only. Rates are PROVISIONAL under gate G9 after a price move and under family F4 after a listing change, so a rate-driven conclusion inside this window is inadmissible by construction.
2. Some losses recover on restoration alone, and a recovery push on top of a self-recovering rank overpays for motion already happening. This sentence is carried forward from v1.0 and it is the entire justification for the step. The test is whether velocity turns positive without spend.
3. Read the daily series against the pre-OOS baseline at the window's close. Three outcomes: rank recovered to baseline, in which case no recovery funds and the term returns to its standing objective; rank moving toward baseline at positive velocity, in which case

the window extends by one read rather than funding a push that would take credit for organic motion; rank flat or still slipping, in which case the term proceeds to P6.

4. AMENDMENT. v1.0 instructed the operator to "observe per the v4.0 observation window for the cause class". No such table exists in v4.0. Family F13 carries persistence classes at syntax grain (TRANSIENT 1-2 weeks, DEVELOPING 3 weeks, CHRONIC 4 or more consecutive weeks) and rule T5 sets daily rank checks for 3 weeks on the post-restock mode specifically. The window used for other cause classes is recorded on the worksheet as an input under test and logged at AM-2. Do not present it as a governed threshold.

5. The out-of-stock class is the exception and it is governed: family F3 states the recovery play triggers at restock confirmation, so P11 deploys at day 0 and this procedure does not run. The drift observed while waiting is the cost of waiting, carried forward from v1.0, and P11 step 1 quantifies it.

6. Where the window closes with rank recovered, score the diagnosis at P14 anyway. A cause correctly named and cheaply resolved is the best outcome this SOP produces and it must enter the calibration series, or the series learns only from the expensive cases.

P6. Recovery classification and ceiling inheritance

Per candidate, 8-12 minutes. Source: v4.0 Section 7 rule D10 and Section 3 family F6.

1. Confirm the class SOP-23 assigned rather than re-deriving it: rule D10 classes every FAR or UNRANKED term RECOVERY where best rank is at or under 10 within the trailing 12 months or the pre-OOS baseline is at or under 10, PROVEN where that rank is at or under 5, and NEVER-RANKED otherwise. Where SOP-23 has not classed the term, read T2 **Best Rank (Hist) + Date** and apply D10 directly and record that this SOP did so.

2. Note the corpus divergence at first use. The KDR Template v3.0 "Bands & Verdicts Reference" sheet defines the recovery zone as "within 10 of Best Rank" and cites it to scenario S-R12. v4.0 Section 3 family F6 defines it as best rank at or under 10 in absolute position, and Section 8.4 carries S-R12 as ADD EXACT + RESTRUCTURE with the RECOVERY-class scenario at S-R13. The identifier half of this divergence is Ledger conflict C17 and v4.0 governs. The definitional half is not C17 and is registered fresh at Section 10.3 as AM-13: "within 10 positions of best rank" and "best rank at or under 10" are different tests that select different terms, and on a term whose best rank was 24 they disagree completely.

3. Inherit the proven ceiling. A RECOVERY-class term skips the cold-start realism penalty and carries the CPC and CVR at which it previously held the position, read from T2 **Pre-OOS Baseline** and the T4 floor register. This inheritance is the reason recovery is cheaper than a cold push and it is also the reason a wrong classification is expensive in both directions.

4. Re-test the O3 clauses that the loss may have moved: indexing and placement per gate G5, purchase intent, CVR at or above benchmark with the source recorded, the competitor-credibility pair (Review Position Gap not FAR-BEHIND, Price Position Index not OUTPRICED), inventory GREEN, and on THIN margin-dollar variations Expected CPC at or under margin dollars divided by 3. Rule S-R13 conditions the recovery push on fundamentals being intact or better against the baseline period, so the comparison is to the baseline state and not to an absolute bar.

5. Where any clause has moved against us since the baseline, the proven ceiling no longer proves anything and the term downgrades to NEVER-RANKED realism for sizing purposes. Record the clause that moved. This is the same downgrade rule S-R13 applies to a failed recovery, applied earlier and more cheaply.

6. Write the class, the inherited ceiling and the re-tested clauses to T2 **Situation** . This record is what SOP-23 reads on any future candidacy for the term, and what P14 scores.

P7. Recovery-adjusted sizing: the three adjustments to the SOP-23 worksheet

Per candidate, 15-20 minutes. Source: rules S-R13 and S-I5, both of which specify M2 sizing against the proven ceiling. The chain itself is SOP-23's and is not restated; only the three deltas are.

1. Assemble the M2 inputs into T2 exactly as SOP-23 does: current rank, target rank or baseline rank, DSTR at the target, validated CVR, Expected CPC, contribution per order from T0 zone C, and current daily orders. Compute order gap, required clicks per day, required spend per day, duration, total budget and the weekly loss ceiling through SOP-23's chain.

2. Adjustment 1, the target. Rules S-I5 and S-R13 both size toward the pre-OOS or best-rank baseline, not toward the standing target rank. Where the two differ, write both into the worksheet and state which the duration was computed against, because the exit tests at P13 read different ones.

3. Adjustment 2, velocity. Recovered terms move faster than cold pushes because relevance signals were retained. AMENDMENT: v4.0 carries no recovery velocity reference. Family F6 bands velocity only in positions per week and supplies no per-class figure, and v1.0's 2.0 positions per day traces to no source. Record the velocity used as an input under test on the worksheet, log it against AM-1, and do not present it as a governed threshold. WE-1 quantifies the exposure: on the same 23-position distance the recovery assumption and SOP-23's cold assumption differ by $1,592.96 of planned spend.

4. Adjustment 3, bid seeding. The previously discovered floor and the prior push CPC bracket the re-entry bid, and the worksheet cites both rather than rediscovering the bid from Expected CPC alone. Read the floor from the T4 register and the prior push CPC from T6. The bracket is a test, not a generator: Expected CPC under rule M2 supplies the seed, and the bracket confirms the seed sits inside the range the term has already demonstrated. A seed outside the bracket is a finding to explain before it is a bid to deploy. AMENDMENT: v4.0 states no rule for choosing a value inside the bracket, and v1.0 wrote one that matches neither endpoint nor the midpoint. On the WE-1 sizing the bracket spans $0.55 , which at 62.96 clicks per day is $34.63 per day and $415.56 across a 12-day

recovery. Logged at AM-7.

5. Run the ceiling read. Compute Max Profitable CPC as contribution times CVR and divide Expected CPC by it. Family F1 bands the ratio at or under 0.8 SAFE, 0.8-1.0 AT-CEILING, 1.0-1.5 PUSH-ONLY, above 1.5 UNJUSTIFIABLE outside a sized push. Above 1.5 inside a sized recovery, rule S-R10 makes the cheap-surface arbitrage check mandatory this cycle at the 0.6 times threshold, and the check is recorded whether or not it fires.

6. Confirm the recovery ACOS sits inside the 50-80% ranking envelope that family F1 sets as the outer band. A recovery outside its objective band is not a smaller recovery; it is a sizing failure that returns to step 1.

7. Apply the elasticity correction where it exists. Rule M2's amendment corrects the click requirement by the term's observed Rank Response Elasticity once 8 or more weeks of push history exist. A recovered term frequently has this history where a cold candidate does not, which is the one place recovery sizing is better evidenced than cold sizing, and it is used wherever available.

8. State what the worksheet could not see, in the same words the engine computed: the provisional margin under D1, an unevidenced DSTR under D8, an account-reference Expected CPC under A6, a baseline CVR standing in for a validated read under G2. Rule J7 makes confidence theater a process violation.

9. Compute the decision confidence score from its three drivers and read it against family F15: HIGH at 75 or above, MEDIUM 50-74, LOW below 50. At LOW, rule M10 bars pushes outright, halves every step and shortens review horizons to one week, and rule D9 limits the legal verdicts to extend-test, data-repair tasks and gate-driven actions. A LOW-confidence recovery does not proceed to P8.

10. Version the worksheet into the T2 plan block. It travels with the recovery, and every later decision cites it by version rather than recomputing it.

P8. Queue submission and the funding decision under product posture

Per candidate, 5-8 minutes. Queue ranking and envelope law belong to the instrument layer; this is the submission and the receipt.

1. Compute the M5 ordering inputs from the worksheet: revenue potential at the baseline rank divided by cost-to-close-gap. Write cost-to-close to T2 **Cost to Close Gap $** . Rule D10 states that RECOVERY terms are funded at better M5 priority, which is a consequence of the inherited ceiling shortening the denominator rather than a separate bonus applied on top; applying it twice double-counts the advantage.

2. Read the product posture before expecting a release. Rule S-A3 grants no new Ranking objectives during a push freeze. Rule S-A4 enters Recovery Mode, where rule R3 permits new Ranking objectives only on RECOVERY-class terms at HIGH confidence, which makes this SOP the single exception to the codered funding freeze and makes the P6 classification and the P7 confidence score the two things that authorize it.

3. Check rule S-A10: a CONSTRAINED earning-potential class bars ranking pushes outright and routes velocity requirements to deals and pricing rather than to a PPC subsidy. Per Ledger conflict C6 the tiering decision sits outside this corpus and this SOP stops at the boundary rather than inferring it.

4. Check gate G11. Where a sibling holds or contests the term, both restore steps hold until #33 declares the lead. Supply this term's recovery class and cost-toclose for the family ordering and execute neither side of the declaration.

5. Submit the worksheet to the queue surfaced in T0 zone H and, for family-level arbitration, Account Rollup tab A2. Per Ledger conflict C3 the submission names the instrument and the tab and never a #31 procedure number.

6. On release, read the dated release and proceed to P11. Without a dated release there is no recovery, and spend moving toward the baseline in its absence is the unfunded-drip condition SOP-23 owns as its F1 and rule S-R11 answers with an immediate ladder toward ceiling.

P9. The pre-OOS taper overlay, run proactively

• Same day on the RED flip, 15-25 minutes. Source: rule S-I3 and family F3. This procedure runs before a loss exists and is the reason most recoveries in this account are cheap. 1. Confirm the trigger: family F3 and rule S-I3 both set it at a projected stockout at or under 14 days on the targeted variation, computed by formula from days on hand, max-sales-day velocity, incoming purchase orders with their arrival estimates and lead time, never eyeballed.

2. Write the pre-OOS baseline snapshot first, before any bid moves. Rule S-I3 requires rank, CVR and CPC per keyword, and family F3 states plainly what the snapshot is for: it is the recovery contract, and a term that stocks out without one is misclassified NEVER-RANKED on its next candidacy and loses the funding priority its proven ceiling earned. Write to T2 **Pre-OOS Baseline** . This is the single highest-leverage step in the document and it costs two minutes.

3. Taper deliberately toward defense rather than stopping. Request the rung values from #28 and stage them; this SOP names the destination and #28 names every step. Rank memory holds better on a taper than on a hard stop, carried forward from v1.0, and the campaign stays live at floor rather than paused for exactly that reason.

4. AMENDMENT. Rule S-I3 states "taper to defense on the variation's terms" and sets no rung count, no interval and no floor for the pre-OOS case specifically. The depth used is recorded as an input under test and logged at AM-10.

5. Re-point ranking campaigns to the declared ranked backup variation the same day per rule L9, with the bid seeded at the backup's Expected CPC. Issue REPOINT VARIATION and hand execution to SOP-12. Where T7 field group D2 has no backup chain for the term, the re-point is not executable and the row is recorded as a blocked verdict class against the D2 manifest row rather than improvised.

6. Pre-build the recovery worksheet during the outage, at P6 and P7, while there is time and while the baseline is fresh. The outage is the cheapest window in the recovery lifecycle to do sizing work in, and a worksheet built at restock is a worksheet built under time pressure against a decaying rank.

7. Log the planned recovery standup against the restock date in T6, with the pre-built worksheet version referenced. Write the taper, the snapshot, the re-point and the planned standup as one dated chain so that the next operator reads a plan rather than four unrelated entries.

P10. The stockout window hold

On stockout despite the taper, 10-15 minutes, then a standing daily check.

1. Rule S-I4 governs: defensive floor only on branded terms, and all push and share objectives to Paused-pending with re-entry criterion "restock confirmed". Issue PAUSED-PENDING (REASON + RE-ENTRY) with both fields populated; a pause carrying only one is incomplete and returns to the pass.

2. Mark the trend windows for exclusion so that the outage does not pollute the four-period reads that will size the recovery. Rule S-I4 requires the marking and gate G10 supplies the mechanism.

3. Stop spend into an unbuyable offer the same day. Advertising a variation that cannot be bought spends into a wall, and the check is buyability rather than eligibility: an offer can read eligible under gate G4 while the buy box is lost or the variation is unorderable. AMENDMENT: v4.0 carries no buy-box or buyability gate, so this test is stated as this SOP's method and logged at AM-11.

4. Read the restock estimate daily against T2 **Days to Stockout** and the inventory feed, and update the planned standup date in T6 on any movement. A standup planned against a stale arrival date is how P11's day-0 discipline fails without anyone deciding to abandon it.

5. Where the outage extends past the point at which the pre-OOS baseline itself ages out of the 12-month window rule D10 reads, flag it: the term is on a clock to lose its RECOVERY class by the passage of time alone, and the finding routes to the inventory owner with the funding-priority consequence quantified.

P11. The post-restock standup at day 0

Same day as the restock landing, 25-35 minutes including staging. Source: family F3 and rule S-I5.

1. Deploy the day stock lands, never after organic drift is observed: the drift is the cost of waiting. Carried forward from v1.0. Quantify it on the record rather than asserting it: at the worksheet's own DSTR and contribution figures, each day of delay is the order gap times contribution per order in contribution not bought, and on the WE-1 sizing that is $95.20 per day . Write the figure into T6 on any delayed standup as the cost of the delay.

2. Confirm the cause is resolved by the landing itself. The out-of-stock class is the one class where P4 restoration and the trigger are the same event, so P5 observation does not run and the recovery deploys directly.

3. Buyability check before deployment: offer live, buy box held, the variation orderable, verified with the artifact rather than assumed from the inventory feed. Per P10 step 3 this test is this SOP's method rather than a v4.0 gate, and it is logged at AM-11.

4. Confirm gate G6 reads GREEN on the restocked variation rather than merely not RED. A restock that lands below lead time plus 21 days of cover at maxsales-day velocity funds a recovery into the next stockout, and family F3's GREEN definition is the test.

5. Re-point back to the primary variation per rule L9, which specifies the reverse re-point running with the recovery play. Stage both the re-point and the recovery bid in one set so they land together.

6. Issue the standup order to SOP-12 carrying five values from the pre-built worksheet and nothing else: the campaign, the re-entry bid the worksheet carries, the daily budget as required weekly budget divided by 7, the objective tag, and the ETA checkpoint date. This SOP writes no bid of its own invention; a creation with no verdict-carried bid returns as incomplete.

7. Confirm the bidding strategy is fixed. Strategic Doctrine v1.0 Section 6.3 sets fixed for ranking pushes because consistency of pressure is the point, and records that the April 2026 platform bidding change weights placement-level conversion probability far more aggressively with category CPC inflation reported at 18-27% against Q1 baselines under dynamic raises.

8. Confirm walls are intact under #29 in the same working session. The original structure's walls may have been altered during the outage, and a recovery whose source campaigns still serve the term buys the same click twice.

9. Exclude the honeymoon-window reads per rule S-I5 and mark the exclusion in T5, so that a post-restock traffic spike is not read as recovery velocity.

10. Set the cadence rule T5 requires: daily rank checks for 3 weeks, with the ETA checkpoint at 3 weeks per rules S-I5 and S-R13. Write the T6 prediction row now, before money moves: metric, direction, magnitude, horizon and review date, which is what P14 scores.

11. Deployment routes through #38 with the T+1 verification pass. The recovery does not count as started until the diff proves it landed.

P12. Compressed in-flight verification

Daily on every live recovery for 3 weeks, 6-10 minutes each. Source: rule T5 and Section 14 rule V4.

1. Read the daily rank series against the worksheet's velocity line and plot the realized positions per day. Where D9 covers the term only weekly, the checkpoint arithmetic carries the coarser interval and the record says so; it does not fabricate a daily read.

2. Read pacing: T2 **In-Budget %** and cumulative spend against the daily line. Below 70% the campaign is truncated under gate G3, the budget lever runs before any bid conclusion, and the week's bid read for the row is void.

3. Read the loss line: cumulative spend minus orders times contribution, against the dated weekly ceiling. Surface the number; the scenario call is v4.0's. 4. Re-read the gates that are also cause classes: eligibility, inventory band, price stability and event mode. Any flip is simultaneously a gate failure and a new cause, and it routes back to P2 rather than forward to a lever.

5. AMENDMENT, and it corrects v1.0. v1.0 instructed that "the day-7 checkpoint becomes day-4" and that a recovery not moving by day 4 is a diagnosis miss. Neither a day-7 nor a day-4 checkpoint exists in v4.0. What v4.0 states is stricter and better sourced: rule T5 sets daily rank checks for 3 weeks and rules S- I5 and S-R13 set the ETA checkpoint at 3 weeks . The v1.0 compression is retired as ungoverned and replaced by the governed daily cadence; the interval at which a flat series becomes a diagnosis miss rather than a patience case is logged at AM-12.

6. No bid changes in this procedure. Pacing interventions touch budget only; bid decisions live in the weekly pass and in verdict-carried ladders. Every intervention writes a T6 line.

7. Run P3 on every read. A confound landing mid-recovery voids the checkpoint rather than failing the recovery, and catching it on the day is far cheaper than reconstructing it at the post-mortem.

P13. The recovery close: taper handoff, renewal, or downgrade

At the checkpoint or on the baseline being reached, 15-25 minutes. Source: rules S-I5, S-R13, S-R16, T1 and T5.

1. Determine which exit test applies, and read both. Rule T5 sets the recovery mode's exit at baseline reached . Rule S-R6 separately sets the taper trigger at 2 consecutive weeks at or under target rank , and rule T1 transitions the objective on the same condition. AMENDMENT: where the pre-OOS baseline is better than the standing target rank, these two tests resolve at different ranks on the same term and v4.0 states no precedence between them. Read both, write both, and route on the stricter until AM-4 resolves.

2. Baseline reached and target held for 2 consecutive weeks: issue TAPER: RANK ACHIEVED, STEP BIDS DOWN and hand to SOP-12 and the #28 ladder. This SOP supplies the achievement date, the recovery bid and the worksheet reference; it supplies no rung, no step size and no floor.

3. Retag the objective per rule T1 and update both the T3 tag and the campaign-name objective segment in the same staged set. The term exits into SOP-25's normalization sequence exactly as an achieved push does.

4. Checkpoint reached with the baseline not held and velocity above zero: this is rule S-R16's renewal decision, never an automatic continuation. Re-size on realized velocity, apply the M5 test to the re-sized figure rather than the original, and renew only where the re-sized cost-to-close still clears priority against the current queue.

5. Checkpoint reached with velocity at or near zero after the required clicks were delivered: rule S-R5's stop-loss conditions are met and the verdict is STOPLOSS: CONCEDE OR DEFER. Confirm all three conditions rather than one, and confirm the top-of-search read at rule S-R7's 30% floor ran first, because a stall at full traffic on the wrong placement is not an auction verdict.

6. The downgrade. Rule S-R13 states that recovery failures downgrade the term to NEVER-RANKED realism, and rule T5 carries the same downgrade as the recovery mode's second exit. Write the downgrade to T2 and to the record SOP-23 reads on the next candidacy, because the whole value of the class is destroyed if a term that has failed recovery twice still presents as carrying a proven ceiling.

7. Where the recovery closes any way other than baseline reached, the freed daily budget line returns to the envelope and reallocates at the instrument layer the same cycle. AMENDMENT: v4.0 states no cool-down before a downgraded term may re-enter the queue; logged at AM-14.

8. Every outcome writes a T6 row with the arithmetic that produced it. A renewal without the re-sized number is not a renewal.

P14. Post-mortem, diagnosis scoring and the serial-recovery escalation

On every closed recovery without exception, 25-35 minutes. Source: Section 14 rules V1 and V5, and SOP-06's rubric.

1. Score the standup prediction HIT, PARTIAL, MISS, or VOID against the horizon it named, and write to T6 **V5 Score** . Where P3 recorded a confound in the read window, the score is VOID and the confound class travels with it; a VOID is not a MISS.

2. Attach the cause tag that #6's rubric requires on every non-HIT: JUDGMENT, EXECUTION, WORLD or DATA. An EXECUTION tag routes to #38 and never to the decider's calibration, which is the point of the axis.

3. Score the diagnosis separately from the prediction. The question is whether the named cause was the real one, and the test is whether rank held after the recovery closed. A recovery that hit its rank and lost it again within the review period scored HIT on the prediction and MISS on the diagnosis, and conflating the two teaches nothing. Write both.

4. Write the three calibration deltas separately: realized velocity against assumed, realized CVR against assumed, and realized cost against budget. Where the velocity delta carries the error, the finding attaches to AM-1, and repeated instances are the evidence that closes it.

5. Feed the realized rank movement per 100 incremental clicks into the elasticity series so that rule M2's correction at 8 or more weeks is available on the next recovery for this term class.

6. Where the recovery failed on a condition a missing feed made unfixable, attach the finding to the T7 row for that field group with the dollar figure attached. A feed with failed recoveries attached to it argues its own priority better than a completeness score of 0.571 does.

7. The serial-recovery test. Read T6 for prior closed recoveries on the same term. Repeated losses on one term are a product, price or competitive problem wearing a PPC costume, and the response is a structural review that routes the four cause records out of PPC entirely, exactly as SOP-26's structural close does. Rule D6 sets the first check on the other side of that boundary: the Price Position Index precedes any listing spend. AMENDMENT: v4.0 states no recovery count and no window; v1.0 asserted three inside "the v4.0 window", which does not exist. The count used is recorded as this SOP's stated method and logged at AM-6.

8. Rule-level accuracy accrues per rule V5: any rule below 50% accuracy over 10 or more scored predictions enters review, and the review produces an amendment, a threshold change or a documented defense. The cause classes are subject to this standard too: a cause class whose diagnoses do not hold is a class that needs re-definition, and Section 1.2 increments on that evidence.

9. A closed recovery with no post-mortem is the F7 condition. Without the diagnosis score the seven cause classes never improve, and the next operator inherits a list nobody has tested.

5. Decision Points: The Full Verdict-to-Execution Mapping

All verdict issuance, thresholds and scenario logic live in v4.0. The closed vocabulary is quoted verbatim from the Keyword Decision Record Template v3.0 "Bands & Verdicts Reference" sheet, which carries the twenty-five strings the exemplar maps, per Ledger conflict C10. This table maps all twenty-five to what this SOP does when each lands on a recovery-relevant row, including the rows where the answer is that another component executes, so that no verdict falls into a seam. Zero local thresholds appear in this table.

Two states the KDR vocabulary cannot express. Per Ledger conflict C18, the KDR twenty-five contains no member for a working recovery continuing unchanged and no member for a family-level lead assignment, and v4.0 Section 13.1 carries both. For those two rows only, the string is quoted verbatim from v4.0 Section 13.1 with the source named in place, exactly as SOP-22 and SOP-23 both do.

Verdict, quoted verbatim	v4.0 Section 13.1 cross-map	Recovery-stage execution
RECOVERY PUSH	No 13.1 member; v4.0 routes S-R13 into RANK PUSH AT CONTROLLED CPC	This SOP's primary verdict. P1 dating, P2 cause, P4 restoration verified with its artifact, P5 observation, P6 classification, P7 the three adjustments, P8 funding, then P11 standup. Issued only on a resolved cause with a verification artifact on file and a RECOVERY or PROVEN class under rule D10. Note the identifier divergence at C17: exemplar SOP-12 v2.0 Section 5 maps this string to S-R12; v4.0 carries it at S-R13 and S-R13 governs.
RANK PUSH: FUND SIZED PUSH	RANK PUSH AT CONTROLLED CPC	Boundary row. SOP-23's verdict, not this one. Its arrival on a term this SOP is diagnosing settles the classification question in the other direction: the term was classed NEVER-RANKED under rule D10, so it carries no proven ceiling to recover toward and no baseline to size against. The diagnosis closes as recorded and the term routes to SOP-23 candidacy. Where this verdict lands on a term that does hold a pre-OOS baseline at or under 10, that is the F6 misclassification condition and the class is re-tested at P6 before any funding.
RE-POINT VARIATION	Identical string	P9 step 5 same day on a RED flip, with the pre-OOS baseline snapshot written before the swap, and the reverse re-point at P11 step 5 on restock. Execution at SOP-12; the backup chain is the declared variation priority flow, and where D2 has no chain the verdict is logged as blocked rather than improvised.
PAUSED-PENDING (REASON + RE-ENTRY)	PAUSE (LEVERS EXHAUSTED)	P10 step 1 on a stockout, with re-entry "restock confirmed" per rule S-I4. Also the verdict on a cause returning UNKNOWN at P2, with the reason naming the untestable class and the re-entry naming the feed that would test it. A pause carrying only one of the two fields is incomplete and returns to the pass.

Verdict, quoted verbatim	v4.0 Section 13.1 cross-map	Recovery-stage execution
TAPER: RANK ACHIEVED, STEP BIDS DOWN	TAPER TO FLOOR (POST-OBJECTIVE)	P13 step 2 on the baseline reached and target held 2 consecutive weeks. Handoff to SOP-12 and the #28 ladder; this SOP names no rung and no floor. Note that the pre-OOS taper at P9 is not this verdict: it is a gate-driven execution under rule S-I3, and v1.0's use of a taper string for it is retired at Section 5.2.
STOP-LOSS: CONCEDE OR DEFER	Identical string	P13 step 5 where all three S-R5 conditions are met and the S-R7 placement read has already run. Concede downgrades the term to NEVER-RANKED realism per rule S-R13; defer requires a new worksheet through the queue. Freed budget returns to the envelope at the instrument layer.
FIX CONVERSION FIRST; DEFER PUSH	Identical string	Where the loss traces to a conversion deficit rather than to a position event. Cap at maintenance, route by funnel drop point with the price check first under rule D6, and set re-entry at CVR back to benchmark. No bid level closes a conversion deficit and no recovery funds against one.
QUARANTINE: DATA REFRESH	FLAG: DATA REPAIR	No loss is dated and no recovery is sized on a quarantined row. Rule P8 makes this beat urgency, which matters here because a stockout in progress creates pressure to diagnose on broken data. The refresh task opens the same day with an owner and a date.
EXTEND TEST	Identical string	The gate G2 outcome this SOP meets most often, because a term that went out of stock has no recent clicks. Below sufficiency the CVR falls back to the 60-day window; where the outage consumed that too, the verdict is EXTEND TEST and the recovery holds. Also the only legal verdict class at LOW confidence under rule D9.
ADD EXACT + RESTRUCTURE	Identical string	Rule S-R12 bars any performance verdict until the exact stands up. A recovering term whose exact was paused during the outage, or which auto and broad are carrying above the X10 share, is a structure repair before it is a recovery. v4.0 Section 16 Worked Pass 3 is the account's live instance: a head term at 74,000 search volume with zero exacts, two autos carrying 78% of its clicks, rank 22 against a target of 5, and a best rank of 7 fourteen months ago pre-OOS. Execution at SOP-12 with steering walls at #29.
PLACEMENT FIX FIRST	No 13.1 member; S-R7 verdict text	Rule S-R7 fires below the 30% top-of-search floor and outranks the triad read. Bids hold; the modifier program belongs to #28. While field group B7 reads MISSING account-wide the modifier half is not executable and the residual is logged as a blocked verdict class on T7.

Verdict, quoted verbatim	v4.0 Section 13.1 cross-map	Recovery-stage execution
DEFEND AT FLOOR	Identical string	The taper-too-deep cause class resolves here rather than in a push: rule M3 restores one step and that value is the recorded minimum viable CPC. WE-4 computes the remedy at $6.60 per week against $392.00 for the sized alternative. The rung value is #28's; this SOP issues the restore instruction.
HOLD; DEFEND LIGHTLY AT RANK	No 13.1 member; the near-neighbour is DEFEND AT FLOOR	The P5 observation window's verdict while a restored cause is under observation: no lever moves, rank is read against the baseline, and the window either closes on self-recovery or routes to P6. Doing nothing is a verdict here and it is logged as one.
CONTINUE; VERIFY MOMENTUM v4.0	Quoted from 13.1 in place	A live recovery tracking its velocity line at P12. No lever moves; the daily check continues and the checkpoint stands. Per C18 the KDR carries no member for this state and v4.0 Section 13.1 is quoted in place with the source named.
PORTFOLIO CALL: ASSIGN LEAD v4.0	Quoted from 13.1 in place	Gate G11 holds both restore steps until #33 declares the lead on a contested recovering term. This SOP supplies each sibling's recovery class and cost-to-close for the family ordering and executes neither side. Per C18, quoted from v4.0 Section 13.1 in place.
BID DOWN TO PROFITABLE CPC; CAP SYNTAX	Identical string	Rule S-R11's answer to spend drifting toward a lost rank with no worksheet: an immediate ladder toward ceiling. A recovery case may be written and sized afterwards, but aggression does not resume before it exists. This is the verdict on the F1 unfunded-recovery condition.
UNLOCK TRAFFIC	Identical string	Rule S-R1 applies post-restock where clicks read below 15 in 7 days with healthy budget and GREEN inventory: incremental bid steps to unlock traffic and no rate judgments until sufficiency. Common in the first days after a re-point, and it is a sufficiency response rather than a recovery verdict.
FIX BUDGET (TRUNCATED)	No 13.1 member; G3 response	Budget lever only at P12 step 2. The week's bid read for the row is void and resumes on the next untruncated day.
TRAFFIC DEFICIT: FEED TO REQUIRED CLICKS	No 13.1 member; S-R3 verdict text	The named lever only, on the dedicated campaign, executed at SOP-12. Never both levers on one row in one cycle. Re-check the ETA in 2 weeks per rule S-R3.
ENTER SBV ARBITRAGE	FORMAT SUBSTITUTION (SBV/SD)	Mandatory check at P7 step 5 whenever a recovery runs above 1.5 times its ceiling. This SOP executes only the SP-exact single ladder step down; the video build belongs to components 17-19.
INCREMENTALITY LADDER	No 13.1 member; v4.0 carries the mechanism at X12	Owned by #35 and by SOP-26 at the mature stage. Listed as a boundary with one live connection: where an incrementality read was riding a window this SOP has voided at P3, the void and its confound references hand to #35 rather than a reading, and #35 re-runs.

Verdict, quoted verbatim	v4.0 Section 13.1 cross-map	Recovery-stage execution
CONSOLIDATE DUPLICATES	Identical string	Boundary row. Owned by SOP-12 P7 with survivor selection human-mandatory. Surfaced here where an outage left a duplicate exact live and unsteered; the recovery does not proceed against split data.
GRADUATE FROM DISCOVERY	GRADUATE + STEER	Boundary row. Owned by #30 and unaffected by a live recovery on an adjacent term. The harvest never stops during an outage or a recovery, and a graduate arriving mid-recovery is stood up and walled the same day.
NEGATE (PRE-LOAD / REACTIVE / STEERING)	CUT / NEGATE	Boundary row. Owned by #29. This SOP names the wall requirement at P11 step 8 and supplies evidence lines; it never selects the wall level.
OPEN / ROTATE / EXIT CONQUEST	Identical string	Boundary row. Owned by #16. Relevant where the competitor-surge cause class identifies a specific holder; this SOP supplies the flag profile and #16 decides whether a counter-open is legal. A recovery is never the vehicle for a conquest response.
CONTINUE TEST TO 50 CLICKS AT CAPPED BID	Identical string	Boundary row. No recovery-stage execution beyond budget assurance. Its arrival on a recovering term means the term is below sufficiency and the recovery has no validated CVR to size against, which routes to EXTEND TEST above.
FUND TEST (UNFUNDED PRIMARY)	FUND DELIBERATE TEST (50-CLICK CAP)	Boundary row. Not issued from this SOP. A term with no rank history is a discovery or cold-push case at #30 or SOP-23, not a recovery: recovery requires a position to return to.
LEAVE TO HARVEST	Identical string	Boundary row. Owned by #30. Its arrival on a term this SOP is diagnosing settles the question: a term the account has decided not to fund an exact for does not receive a recovery push, and the diagnosis closes as recorded rather than as actioned.

5.1 Campaign-level verdicts

Campaign rows carry their own subset vocabulary, quoted from the KDR Template v3.0 "Campaign Decision Record" tab: FUND / HOLD / TRIM BUDGET, RAISE / CUT MULTIPLIER, NEGATION PASS, CONSOLIDATE, RETAG OBJECTIVE, PAUSE (L7), ARCHIVE. On a recovery, RETAG OBJECTIVE is the verdict that catches a campaign still tagged Ranking after a recovery closed into the taper, which is failure mode F11. PAUSE (L7) never applies to a pre-OOS campaign, because P9 step 3 keeps the campaign live at floor deliberately.

5.2 Verdict strings retired at this version

SOP-24 v1.0 Section 5 issued four strings that appear in neither the KDR Template v3.0 closed vocabulary nor v4.0 Section 13.1: **RANK PUSH: RECOVERY PUSH** (a concatenation of two separate KDR members), **HOLD FOR OBSERVATION** , **TAPER (PRE-OOS)** , and **RETURN TO DIAGNOSIS** . A closed vocabulary is what makes the decision log queryable and predictions scoreable per verdict type under rule V5, and a locally minted string is invisible to both. All four are retired. The operations they named survive as procedures rather than as verdicts: the recovery push is RECOVERY PUSH quoted from the KDR, the observation window is P5 and issues HOLD; DEFEND LIGHTLY AT RANK, the pre-OOS taper is P9 and issues RE-POINT VARIATION with a gate-driven ladder instruction to #28, and the return to diagnosis is P12 step 7 and P2 re-entry, which is a routing rather than a verdict.

6. Worked Examples

All four run on the standard training base, disclosed here because live per-variation economics are not attached to this session: AOV \$80.00, contribution \$28.00 per order, break-even ACOS 35.0%, Target ACOS 17.5%, Max Acceptable 26.3%. This is external dependency D1 in the Ledger, open, and it caps this component by name. Every figure below was computed and asserted in code with float tolerances before it was written here. WE-1's DSTR anchor is carried unchanged from the Economics Worked-Example Set case E3 and WE-4's floor anchor from case E5, so that the documents cannot drift apart.

WE-1 · A clean post-OOS recovery, and the three governance gaps it exposes

Setup. The king variation goes out of stock for 9 days. The head term held organic rank 7 before the outage and reads 31 at restock. Standing target rank is 8. This is the v1.0 incident, recomputed at record grain on the training base.

P9, run during the outage. At the RED flip the pre-OOS baseline snapshot was written: rank 7, CVR 5.4%, CPC $2.40. The campaign tapered toward defense and stayed live at floor $1.85 rather than pausing. The worksheet was pre-built while the baseline was fresh.

P1 and P2 at restock. The daily rank series inflects on day 1 of the stockout window, inside it rather than before it, so the OOS class is admitted. T5 carries no price, listing or competitor event in the window. Primary cause: OOS, evidence attached. P4 restoration is the landing itself; P5 does not run, because family F3 ties the recovery play to restock confirmation.

P6 classification. Pre-OOS baseline rank 7 is at or under 10, so rule D10 classes the term RECOVERY. It is not at or under 5, so it is not PROVEN. The proven ceiling is inherited from the baseline CVR of 5.4%.

P7 sizing, with the three adjustments.

Distance to target: 31 minus 8 = 23 positions

DSTR at rank 8 = 4.0 orders/day (E3 anchor); current daily orders 0.6; order gap = 3.4 Required clicks/day = 3.4 / 0.054 = 62.96

Bid seeding: bracket [floor $1.85, prior push CPC $2.40]; Expected CPC $2.30 falls inside, so the seed stands at $2.30 Required spend/day = 62.96 x $2.30 = $144.81 ; required budget = $1,013.70/wk

Duration = 23 / 2.0 positions per day = 11.5, planned 12 days (velocity is an input under test, per AM-1) Total recovery budget = $144.81 x 12 = $1,737.78

Paid orders/day = 62.96 x 0.054 = 3.4; contribution recovered = 3.4 x $28.00 = $95.20/day Expected loss/day = $144.81 minus $95.20 = $49.61 ; over 12 days = $595.38

The ceiling, computed by the M2 formula rather than estimated. Recovery ACOS = $144.81 / (3.4 x $80.00) = 53.24% , inside the 50-80% ranking envelope. ACOS above break-even = 53.24% minus 35.0% = 18.24 points. Projected weekly sales = 3.4 x 7 x $80.00 = $1,904.00. Weekly loss ceiling = 0.1824 x $1,904.00 = $347.30 , which across the 12-day plan is $595.38 and reconciles exactly to the expected-loss total above. Ceiling read: Max Profitable CPC = $28.00 x 0.054 = $1.51; $2.30 divided by $1.51 is 1.52 times ceiling , above the 1.5 line, so rule S-R10 makes the cheap-surface check mandatory this cycle at 0.6 x $2.30 = $1.38. Recorded either way.

What the RECOVERY class is worth, and why AM-1 matters. Sized cold at SOP-23's 1.0 position per day, the same 23-position distance runs 23 days at $144.81, a total of $3,330.74 with expected loss of $1,141.14. The recovery classification therefore saves $1,592.96 of planned spend and $545.76 of planned loss on one term. That entire saving rests on a velocity number v4.0 does not carry. The classification is governed; the figure that converts the classification into money is not.

P12, the daily reads. Rank moves 31 to 24 by day 4, which is 7 positions in 4 days, or 1.75 per day against the 2.0 line. v1.0 called this the day-4 checkpoint and treated a flat read here as a diagnosis miss. There is no day-4 rule in v4.0. What v4.0 states is stricter: rule T5 requires daily rank checks across the full 3 weeks, so this read is one of twenty-one rather than a gate, and the checkpoint sits at day 21 per rules S-I5 and S-R13.

Day 12 and the exit-test collision. Rank reaches 8 on day 12, realized velocity 23 / 12 = 1.92 positions per day , 95.8% of the assumption. Two exit tests now apply and they do not agree. Rule T5 exits the recovery mode at baseline reached , and the baseline is rank 7, not 8. Rule S-R6 fires the taper at 2 consecutive weeks at or under target rank , and the target is 8, which is now held. On this term the mode exit and the taper trigger sit one position and fourteen days apart, and v4.0 states no precedence between them. This is AM-4.

What the stability window costs. Holding levers still for 14 days at recovery pressure spends a further $2,027.41 and accrues a further $694.61 of expected loss. Cumulative loss by day 26 reaches $1,289.99 . The weekly ceiling of $347.30 scales with time, so this is not a ceiling breach; it is worse in a quieter way. Rule M2 requires the ceiling to be capped and dated , and the dated horizon was day 12. The window runs 14 days past that date against a ceiling nothing re-dates, which means the spend is not over its limit, it is outside the instrument that sets one. This is AM-5.

Close. On day 26 the taper issues, handing off to SOP-12 and the #28 ladder. P14 scores the prediction HIT at day 12 against a day-12 horizon. Diagnosis scored correct: rank held after the recovery closed. The three deltas: velocity 1.92 realized against 2.00 assumed, CVR 5.38% against 5.40%, cost $1,681 against $1,737.78 budgeted. The velocity evidence accrues against AM-1.

WE-2 · The edge case: four days of dating overturn the obvious cause, and the recovery must not fund

Setup. A head term held rank 6 for eleven weeks and reads 22 across the last three. The product log shows a 6-day stockout, so the operator's first read is OOS and the pre-built worksheet is already in the folder from a prior cycle. The stockout was on a secondary variation, not the targeted one.

P1 dating, which is the whole example. D9 covers this term at daily resolution because it is a head term, so the inflection date is exact. The series leaves its held band on the Tuesday; the stockout window opens on the Saturday. The inflection precedes the stockout by 4 days. Under P2 step 2 the direction of that comparison rules the OOS class out rather than in, and the pre-built worksheet is now a worksheet for the wrong cause.

P2, in the order rules D5 and D6 require. Price first. Our price reads $79.95 and is unchanged in T5. The market limb requires the SQP median market price paid, and T7 field group D3 reads MISSING , so the Price Position Index cannot be computed and the class cannot be tested. Listing: no events in the window, gate G5 passes, index confirmed. Review: Review Position Gap reads PAR, which neither admits nor excludes. Competitor surge: the class that fits the shape of the loss, and T7 field group D8 reads MISSING , so it is evidenced only if someone logged a competitor entry to T5 by hand, and nobody did. Deal hangover: no window. Taper too deep: T6 carries no ladder step inside the window and the #37 rule change log is clean.

The verdict, and the discipline it takes. Two of seven classes are not testable and five are tested and negative. Under P2 step 7 the record reads UNKNOWN with "not testable" written against price and competitor surge, never "not found", because the two are a manifest finding and a diagnostic conclusion respectively. The verdict is PAUSED-PENDING (REASON + RE-ENTRY), reason "cause UNKNOWN; D3 and D8 unavailable", re-entry "Price Position Index computable and at or better than PAR, or a dated competitor event in T5".

What the wrong call would have cost. Had the pre-built OOS worksheet funded:

Distance 22 to 6 = 16 positions; DSTR at rank 6 = 5.0/day; current 2.1; order gap = 2.9 Required clicks/day = 2.9 / 0.048 = 60.42 ; Expected CPC $2.60; required spend/day = $157.08 Duration at 2.0 positions/day = 8 days ; total = $1,256.67

Contribution recovered = 2.9 x $28.00 = $81.20/day; expected loss = $75.88/day , $607.07 over the plan

Recovery ACOS = $157.08 / (2.9 x $80.00) = 67.7%, inside the envelope; Max Profitable CPC = $28.00 x 0.048 = $1.34; CPC ratio 1.93 times ceiling

Every one of those figures is internally sound. The sizing chain was not wrong; it was pointed at a cause that had already been ruled out by four days on a rank series. If the true cause is a competitor price cut that has not reverted, the rank the $1,256.67 buys is taken back the week the recovery stops, and the term returns to the queue the next quarter presenting as a fresh candidate with its RECOVERY class intact and one more failed recovery in its history that nobody reads.

Where the finding goes. Attached to the T7 rows for D3 and D8, with $1,256.67 of held recovery attached to it. A price feed and a competitor feed with a blocked diagnosis and a quantified candidate attached argue their own priority more effectively than a completeness score of 0.571 does. This is the failure gate G8 exists to make visible: not a confident wrong answer, but a correctly withheld one whose cost is invisible unless someone writes it down.

WE-3 · The confound: an event window suppresses the read the checkpoint requires

Setup. A bamboo head term, restocked and recovering. Rank 25 at standup against a pre-OOS baseline of 6. Standup day 0, all gates clean. Distance 25 to 6 = 19; DSTR at rank 6 = 5.0/day; current 1.4; order gap = 3.6

Required clicks/day = 3.6 / 0.052 = 69.23 ; Expected CPC $2.15; required spend/day = $148.85 Duration = 19 / 2.0 = 9.5, planned 10 days ; total = $1,488.46

Contribution recovered = 3.6 x $28.00 = $100.80/day; expected loss = $48.05/day

Recovery ACOS = 51.7%, inside the envelope. Max Profitable CPC = $28.00 x 0.052 = $1.456; CPC ratio 1.48, PUSH-ONLY band, so S-R10 does not fire

Day 9: a deal window arms on the product. The window runs days 9 to 13. Gate G10 excludes event weeks and the 14 days following from trend verdicts, so the guard runs to day 27 . Rules S-I5 and S-R13 set the ETA checkpoint at 3 weeks, which is day 21 .

The collision, stated plainly. The checkpoint the recovery is governed by lands six days inside a window in which the read that would fire it is inadmissible. P3 runs and returns VOID rather than MISS: the confound is documented, its class is "event or deal window under gate G10", and the score files as VOID with the confound references attached. Rule S-E5 governs the other half: had the recovery touched the baseline inside the window, the achievement would be provisional and the two-week at-target clock under rule T1 would start only after the guard closes, so no taper begins on the event spike.

What the void costs, and why it is an amendment rather than a judgment call. Re-dating the checkpoint to the guard end runs the recovery a further 6 days at $148.85, which is $893.08 of spend and $288.28 of expected loss that no rule sizes and no scenario authorizes, because rule S-R13 sets a checkpoint and gate G10 suppresses the read that fires it, and v4.0 states no re-dating rule for the suppressed interval. Deeming the recovery failed instead would be worse and equally ungoverned: it would file a MISS against a sound assumption and teach the calibration series the wrong lesson. The extension is recorded as an input under test and logged at AM-5.

The boundary the example also settles. When the window closes on day 13 the term dips again inside the settle. That dip is not a new loss and does not open a second recovery: per the Ledger row for #32, a post-deal dip inside settle is a hangover resolving by calendar, and family F13 classes it TRANSIENT at 1-2 weeks. P2 step 6 stops there. The recovery already live continues; a second one does not open. If the depression survives settle expiry, P1 dates it fresh from that date.

Hand-off. The void and its class references go to #6 for the score and to #35, which had an incrementality read riding the same two-week window. #35 reruns outside the confound. A confounded read handed over as clean is worse than no read, because the receiving component cannot detect the contamination.

WE-4 · Serial recovery, the taper-too-deep class, and the 59-times error

Setup. One term, three closed recoveries inside the review period. Loss 1 was the WE-1 stockout at $1,737.78 . Loss 2 was diagnosed as a competitor surge and recovered at $980.00 . Loss 3 was diagnosed the same way and recovered at $1,420.00 . Cumulative recovery spend on one term: $4,137.78 . All three recovered their rank, so all three scored HIT and none was ever reviewed.

What P14 finds on the third pass. Reading T6 for prior recoveries surfaces the pattern the individual post-mortems could not see. Re-dating loss 2 against T6 rather than against T5 shows the rank series inflecting on a dated ladder step: the discovered floor was $2.24 and the quarterly probe stepped it to $2.02 , a 9.8% step inside the M3 size and inside the L2 per-change limit and above the L2 floor. Rank slipped 4 to 9. The cause class was never competitor surge; it was taper too deep , and it was ours.

The arithmetic that makes the class worth having.

Correct remedy, rule M3: restore one rung, $2.02 back to $2.24, a 10.9% change inside the L2 limit

Cost at 30 maintenance clicks per week: ($2.24 minus $2.02) x 30 = $6.60 per week

What was done instead: a sized recovery on the 5-position slip. DSTR at rank 4 = 4.4/day; current 3.1; order gap 1.3

Required clicks/day = 1.3 / 0.052 = 25.0 ; at $2.24 that is $56.00/day , or $392.00 per week Difference: $392.00 minus $6.60 = $385.40 per week , a factor of 59.4

Annualized on one term at one misread class: $20,040.80

Why the misread is easy. A taper-too-deep loss and a competitor-surge loss produce the same rank series and the same T5 log window, which is empty in both cases. The only artifact that separates them is the T6 ladder date, and T6 is the one log an operator diagnosing a rank loss has no instinct to open, because it records what we did rather than what happened to us. That is why P2 step 6 puts the T6 read and the #37 rule change log last in the sequence but inside it, and why the empty-log finding at P1 step 7 points at exactly these two classes.

The escalation. Three losses on one term is a product, price or competitive problem wearing a PPC costume, and the four cause records route out of PPC to the cross-framework read exactly as SOP-26's structural close does. Rule D6 sets the first check on the other side: the Price Position Index precedes any listing spend. The count itself is not governed. v1.0 asserted "the third recovery inside the v4.0 window"; v4.0 carries no recovery count and no window. The count used is this SOP's stated method and is logged at AM-6, with this case as the evidence that closes it.

The correction that survives the escalation. Independent of where the structural review lands, loss 2's floor register entry is wrong: a probe that failed does not lower the register, so the recorded floor returns to $2.24 and the $2.02 rung is marked as a failed probe with its date. The register is #28's and this SOP supplies the correction rather than making it.

7. Failure Modes and Escalation

Thirteen modes. Each carries the signal that detects it, the response, and the account incident that motivated it where one exists. The five incidents recorded in v1.0 carry forward here because they cannot be reconstructed from any other document in the library, and they are marked.

ID	Failure	Detection signal	Response	Motivating incident
F1	Recovery deployed before the cause is resolved	A recovery standup in T6 with no P4 verification artifact filed against it, or with a cause class of UNKNOWN.	Blocked at the gate. The standup reverts and the term holds at PAUSED-PENDING until the artifact exists. A recovery fired before the cause is resolved buys rank the cause takes back.	v1.0 The artifact rule exists because this happened: v1.0 records the verification requirement as its own F1, which means the pattern preceded the rule. WE-2 is the live shape of it, with a pre-built worksheet ready to fund against a ruled-out cause.
F2	Waiting out the restock	Restock landed in the inventory feed, no standup deployed, drift accruing on the daily rank series.	P11 day-0 rule. The drift days are quantified in the T6 note as the cost of the delay: order gap times contribution per order per day, which on the WE-1 sizing is $95.20 per day.	v1.0 Recorded as v1.0's F2 with the drift-cost quantification already prescribed. The doctrine carried forward with it: the drift is the cost of waiting.

ID	Failure	Detection signal	Response	Motivating incident
F3	Cold sizing on a recovery	A recovery worksheet using the cold velocity assumption, or one that ignores the inherited ceiling and re-derives the bid from Expected CPC alone.	P7's three adjustments applied. Cold sizing overpays duration and budget: on the WE-1 distance the difference is $1,592.96 of planned spend.	v1.0 Recorded as v1.0's F3. The correction at v2.0 is that the recovery velocity reference it prescribed does not exist in v4.0, so the mode now detects cold sizing but the corrected figure is an input under test at AM-1.
F4	Serial recovery without escalation	T6 carrying repeated closed recoveries on the same term inside the review period, each scored HIT individually.	Structural review at P14 step 7; the cause records route out of PPC. Three successful recoveries look like three wins and are one unresolved problem.	v1.0 Recorded as v1.0's F4 with the doctrine line carried forward verbatim in substance: three losses is a product, price or competitive problem wearing a PPC costume. WE-4 computes an instance at $4,137.78.
F5	Spend continuing into an unbuyable offer	Live spend on a variation that is out of stock, suppressed, or has lost the buy box.	P10 step 3, same day, with the spend quantified. Escalates on first occurrence: this is the emergency class of this document, because the money buys nothing at all rather than buying badly.	v1.0 Recorded as v1.0's F5, which already required same-day escalation with the spend quantified.
F6	Recovery misclassified as a cold start	A term with a best rank at or under 10 in the trailing 12 months carrying NEVER-RANKED realism, or a stockout closed with no pre-OOS baseline written.	Re-class under rule D10 and re-size. Where the baseline was never written the class cannot be restored from evidence and the term funds at cold priority, which is the permanent cost of a missed two-minute step.	v4.0 Section 3 family F3 records the motivating case directly: a rank-80 term that held rank 6 before the stockout is a RECOVERY candidate with a proven ceiling, not a never-ranked hopeful. v4.0 Section 16 Worked Pass 3 carries a live instance at best rank 7 fourteen months ago.
F7	Recovery closed with no post-mortem or no diagnosis score	A closed recovery in T6 with no V5 score, or with a prediction score and no separate diagnosis score.	P14 runs late. Without the diagnosis score the seven cause classes never improve and the next operator inherits a list nobody has tested.	v4.0 Section 14 rule V1 states the principle: unscored actions do not accumulate. SOP-06's rubric requires the cause tag on every non-HIT and cannot compute without it.
F8	A confound filed as a MISS	A non-HIT score with a documented event, deal window or price test inside its read window, and no VOID.	Re-score as VOID with the confound class attached, and re-run the read outside the window. Also route the correction to #6 and to #35 where either consumed the wrong score.	SOP-06's rubric names this as its own F1 outcome-bias class and names VOID abuse as its F3, which is the same error in the other direction. This SOP publishes the classes that make both detectable.

ID	Failure	Detection signal	Response	Motivating incident
F9	Diagnosis on a stale or unavailable rank series	An inflection date written without its interval, on a term outside D9's head-term coverage.	The loss is not dated and P2 does not run. The finding attaches to the T7 D9 row rather than resolving into a weekly-resolution guess presented as a daily one.	The Data Architecture records a live instance of the class: the first cycle logged a 126-against-101 rank disagreement between sources and correctly filed it as a flag rather than picking one.
F10	Recovery funded during a push freeze or outside Recovery Mode's exception	A recovery release dated inside an S-A3 push freeze, or inside Recovery Mode on a term that is not RECOVERY-class at HIGH confidence.	The release reverts. Rule R3 makes RECOVERY-class terms at HIGH confidence the single exception to the code-red funding freeze, and it is an exception with two conditions, not one.	v4.0 Section 3 family F2 records the account's standing state: roughly 20% blended TACOS against mature-stage products is a standing CODE RED, which makes this exception live rather than theoretical on multiple products.
F11	Objective retag missed after a recovery closes	T3 objective tag reading Ranking against a campaign-name objective segment or a T2 state that has moved to post-objective.	Retag both surfaces in one staged set. A recovered term still tagged Ranking generates a false finding every cycle and reads as a live push to every rollup that counts them.	Shared with SOP-12's F7 and SOP-26's F5, which records push-era structure surviving into the mature stage as a standing account pattern.
F12	Pre-OOS baseline never written	A variation that entered RED with no T2 **Pre-OOS Baseline** row, detected on the next candidacy rather than at the time.	Standing audit line on the weekly pass for every variation in RED or YELLOW. Rule S-I3 requires the snapshot and P9 step 2 places it before any bid move for exactly this reason.	SOP-23's F2 records the account instance: a head term fell from rank 7 across a stockout and was later mistaken for a cold start, which is the misdiagnosis the snapshot exists to prevent.
F13	A restored rung read as a market loss	A rank slip whose date matches a T6 ladder entry or a #37 automated step, diagnosed as competitor surge or as an unclassified loss.	Re-class to taper too deep and restore one rung per rule M3 rather than sizing a push. Correct the floor register: a failed probe does not lower the register.	WE-4 computes the account cost of the class at $385.40 per week and $20,040.80 annualized on one term. SOP-26's F11 records the automated version, where one rule stepped a defensive term six rungs against a stale target and nothing alarmed because each step was inside its own limit.

Escalation. F5 escalates same day on first occurrence with the spend quantified, because advertising into an unbuyable offer is the only mode here that returns nothing at all. F1 and F10 escalate on discovery because both corrupt a funding decision. F4 escalates by definition, since the structural review is itself an escalation out of PPC. Any mode occurring twice in consecutive cycles on the same scope escalates to the PPC Lead with the T6 history attached. Everything in the human-mandatory tier is confirmed per the Authority Matrix before deployment, without exception and without a Recovery-Mode discount.

8. Outputs and Artifacts

Gates clear with artifacts rather than assertions. Each row names the record, where it is stored, how long it is kept, and who consumes it next.

Artifact	Produced by	Storage and retention	Downstream consumer
The attribution statement: inflection date, its interval, the position-class transition, and the log window	P1	T2 **Situation** with the cycle date; retained through the term's next candidacy.	P2, which cannot run without it; the next cycle's P1, so a repeat loss is dated against the prior one rather than from scratch.
The cause record: one primary class with its evidence, contributing conditions, and every class marked not testable	P2	T2 **Situation** and T5 as a dated event; retained for the life of the term.	SOP-06, which defers its confound cause classes here by name; #35, for its void classes; SOP-25, whose descent-slip diagnosis reads them; the diagnosis engine, which records this as one of its two entry points.
P3			
The void determination with its confound class		T6 against the affected read, with the guard dates computed.	#6 for the score, which files VOID rather than MISS; #35 where an incrementality read shared the window.
The restoration verification artifact	P4	T5 as a dated event with the artifact referenced; T6 for the ladder-restore case.	P11, which will not deploy without it; the F1 audit line; the listing, pricing and review owners where the restoration was theirs.
The pre-OOS baseline snapshot: rank, CVR and CPC per keyword	P9 step 2	T2 **Pre-OOS Baseline**; retained at least the trailing 12 months that rule D10 reads.	P6 classification on this recovery and on every future one; SOP-23's candidacy classification; the D10 read that determines funding priority.
The pre-built recovery worksheet, versioned	P7	T2 plan block; superseded versions kept for calibration.	The instrument-layer M5 ordering; #28 for the authorized CPC envelope its ceiling worksheet tests against; P14 for the calibration deltas.
The queue submission with cost-to-close	P8	T0 zone H; Account Rollup tab A2 for family arbitration.	The M5 queue; #33 where the term is contested, which reads the recovery class as an ordering input.
The standup record with its prediction	P11	T6 Decision Log, append-only.	SOP-12 for execution; #38 for staging and the T+1 diff; P14, which scores the prediction it carries.
Daily rank and pacing lines	P12	T6, one line per intervention; alarm evidence retained through the close.	P13's checkpoint arithmetic, which inherits the series rather than re-deriving it; the F9 coverage audit.
The close record: taper handoff, renewal arithmetic, or downgrade	P13	T6 with the realized figures; the downgrade written to T2.	SOP-12 and #28 on a taper; SOP-25 on the post-objective handoff; SOP-23, which reads the downgrade on the next candidacy; the instrument layer on returned budget.
The post-mortem: prediction score, cause tag, separate diagnosis score, and three calibration deltas	P14	T6 with the V5 score; deltas retained indefinitely as the elasticity and cause-accuracy series.	P7 on the next recovery for this term class; rule V5 rule-level accuracy; Account Rollup tab A3; the training corpus under rule V8; the T7 row for any feed a failure was traced to.

Gate artifacts. Before funding, the restoration verification artifact is the gate artifact and no recovery deploys without one. In flight, the daily rank line is the standing artifact and its absence means the recovery was not operated. At close, the post-mortem with its separate diagnosis score is the gate artifact and a recovery cannot be marked closed without one. Retention: no rule in v4.0 sets a retention period for any of these records. The rows above state the log's own append-only life, which is a property of the instrument rather than a governed period. The gap is AM-15 and it duplicates items logged by prior build sessions, so v4.1 folds it once.

9. Skill-Conversion Block

Block	Specification
Input bindings	By catalog ID: A1 four-period comparability, A3 pacing and in-budget, A4 sufficiency buckets, A5 threshold flags, A6 objective and per-term Expected CPC, A7 eligibility, B1 indexing and listing placement, B3 rank pack, B8 economics chain, B11 market context, D1 per-variation margin, D2 keyword-variation map and backup chains, D3 SQP median price, D6 sibling view, D7 decision log and V5, D8 competitor depth, D9 daily rank series. Field-level: T2 rank block for **Best Rank (Hist) + Date**, **Pre-OOS Baseline**, **OOS Overlay Flag** and **Velocity (pos/wk)**; T2 gates block for **Inventory Band**, **Days to Stockout**, **Eligibility**, **Indexed?** and **Event Mode**; T2 plan block for the worksheet; T2 economics block for ceilings; T4 **Ladder State (step / date)** and **Discovered Floor $**; T0 zone C for the spine and zone H for the queue; T5 for the product log, which is this skill's primary evidence surface and the one no other skill in the library depends on this heavily; T6 for predictions, ladder dates and scores; T7 for the manifest. The #37 rule change log for the automated half of the taper-too-deep class. An absent feed suppresses its dependent cause class and the skill writes "not testable" rather than "not found"; it never improvises around a missing feed and never collapses a manifest gap into a diagnostic conclusion.
Deterministic logic	The inflection search across the daily series with its resolution interval propagated into every downstream date comparison. The seven-class decision tree run in the fixed order rules D5 and D6 impose, resolving to exactly one primary class, one or more contributing conditions, or UNKNOWN. The directional stockout test (inflection inside the window admits, before the window excludes). The G10 guard-end computation and the VOID determination with its five confound classes. The D10 classification and the ceiling inheritance. The M2 chain with the three recovery adjustments, the bracket test on the seed, the four CPC-ratio bands, and the confidence bands with their step modulation. The dual exit test at P13, evaluated against both the baseline and the target with the divergence stated. The serial-recovery read across T6. Scenario selection, verdict issuance and every threshold are v4.0's; the skill version-pins v4.0 and re-validates Section 5 on any increment before running a pass. No local thresholds.
Human checkpoints Drawn from PPC Authority Matrix v2.0, Section 1, by row	Executable with audit: loss dating and the inflection search, cause-evidence assembly, void determination, drift-cost quantification, pre-built worksheet computation during an outage, the daily pacing and rank lines, the re-sized renewal arithmetic, and the post-mortem deltas. Human-mandatory: the cause confirmation and the restoration sign-off, because a wrong cause is the failure this document exists to prevent and no arithmetic detects it; the funding decision per the matrix row "Push funding (the M5 queue)", where the CEO decides above the envelope and the PPC Lead within it; the standup per "Structure creation (all SOPs)"; every deployment confirm per "Bulk deployments (#38)"; the shared-term restore step per "Sibling declarations (S-series)"; the downgrade to NEVER-RANKED realism and the F4 structural escalation, both of which change how a term is treated for a year. Weekly recovery verdicts sit on the matrix row "Weekly verdicts (established US)": the PPC Lead decides, the operator executes assigned scope, and the brand-management owner reviews in the daily huddle, which is review rather than veto with disagreements routing up rather than sideways. Under the vacancy rule an unfilled role's decision right reverts to the PPC Lead and its execution to the PPC Manager. Recovery Mode tightens confirmation thresholds per rule R3 and the skill reads the tightened values while the mode is live.
Cadence	Daily: the rank-loss scan on every held term, the pre-OOS trigger watch on every ranked term's variation, and in-flight rank and pacing checks for 3 weeks on every live recovery per rule T5. Same day: the RED flip and the P9 overlay, the stockout hold, the restock standup at day 0, eligibility flips, and product posture changes. Weekly: cause classification and the observation-window reads inside the Monday cascade, deployment Tuesday per #38, T+1 verification Wednesday. At the 3-week checkpoint: the P13 close. On close: the post-mortem. Per review period: the serial-recovery read across T6.

Block	Specification
Alarm specification	F1 to F13 each carry a detection signal from Section 7, an evidence-line format of row identifier, signal value, the v4.0 rule or gate reference, and the artifact where the signal was read, plus a routing tier. Same day to the operator queue: F1, F5, F10, F12. Into the weekly pass worklist: F3, F8, F9, F11, F13. To the PPC Lead with T6 history attached: F2, F4, F6, F7, and any mode recurring in consecutive cycles.
Outputs	Drafted attribution statements with their dating intervals; drafted cause records with evidence and explicit not-testable marks; void determinations with their confound classes; pre-built worksheets with their manifest-gap statements; drafted standup packets; daily rank and pacing lines with drift-cost quantification on any delay; drafted close records with the dual exit test evaluated; drafted post-mortems with the prediction score and the diagnosis score separated; and F1 to F13 alarms with evidence lines. Everything is drafted and queued for confirmation at the tier the matrix names; nothing in the human-mandatory set deploys from the skill.

10. Version Governance

Owner: PPC Lead, with the brand-management counterpart per product on the listing, price and review cause classes, which resolve outside PPC.

10.1 Review triggers specific to this component

• Any v4.0 increment touching rule D10, rules S-I3, S-I4 or S-I5, rule S-R13, rule S-R16, rule T5, rule M3 or gate G6. Each is load-bearing in a named procedure, and the Section 5 mapping re-validates against the new verdict set before any pass runs.

- • Any amendment item in Section 10.3 landing in v4.1. AM-1 in particular converts P7 step 3 from an input under test into a governed threshold, which changes every recovery sizing in the account, and AM-4 converts P13 step 1 from a dual read into a single test.
- • Any confirmed new cause class. The list at Section 1.2 is closed at seven and is consumed by name by SOP-06, #35 and SOP-25, so an increment here propagates to three documents and this SOP notifies all three on publication.
- • Any change to SOP-23's sizing chain, since this document inherits it by reference and states only the three deltas.
- • Any change to SOP-23's sizing chain, since this document inherits it by reference and states only the three deltas. • Any change to #28's ladder mechanics, the reversion rule or the floor register schema, since the taper-too-deep class, the P9 overlay and the P13 handoff all execute against them.
- • Landing of field group D3 SQP median price, which converts the price cause class from half-testable to standing; of D8 competitor depth, which does the same for competitor surge; and of D9 at full coverage, which converts loss dating from a head-term capability into an account-wide one.
- • External dependency D1 landing. The per-variation margin table removes the PROVISIONAL tag from every ceiling here and makes the worked examples reproducible on live economics rather than the training base.

Quarterly at minimum, alongside the floor re-tests that rule V4 places on that cycle.

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10.2 Change note, v1.0 to v2.0

Supersedes v1.0 in full. Procedures expanded from 5 to 14: loss dating, cause classification and the confound determination were compressed inside v1.0's single P1 and are now three separately executable procedures, because SOP-06 and #35 consume the second and third independently; restoration and observation split out of v1.0's P2; classification and sizing split out of v1.0's P3; the pre-OOS overlay, the stockout hold and the post-restock standup split out of v1.0's P4; and in-flight verification, the close and the post-mortem are new. Verdict mapping expanded from 5 rows to the full 25-verdict closed vocabulary with boundary statements, plus the two states v4.0 Section 13.1 carries that the KDR does not, per Ledger conflict C18. Worked examples expanded from 1 to 4. Failure modes expanded from 5 to 13, with all five v1.0 modes and their incidents carried forward. Elements 11 and 12 added, neither having any model in the corpus.

Six corrections of substance. First, the v1.0 header citation of "v4.0 Section 8.2" is corrected: Section 8.2 is Syntax-Level Scenarios and carries no recovery content; the authorities are Sections 8.3 (S-I5), 8.4 (S-R13) and 11 (T5). Second, three v1.0 numerals attributed to v4.0 and not present there are retired rather than carried: the 2.0 positions-per-day recovery velocity reference (AM-1), the day-4 checkpoint (AM-12), and the third-recovery escalation count (AM-6). Third, four locally minted v1.0 verdict strings are retired at Section 5.2. Fourth, the v1.0 citation of "#31 / P1.4" for out-of-stock forecasting is retired to the instrument layer per Ledger conflict C3; the Ledger names it as one of the five dangling citations that conflict records, and it is closed at source here. Fifth, v1.0's instruction to "observe per the v4.0 observation window for the cause class" is retired because no such table exists (AM-2). Sixth, the definitional divergence in the recoveryzone test is registered fresh at AM-13, distinct from the identifier divergence already registered as C17.

Carried forward. All seven v1.0 cause classes, now sourced to v4.0 rules rather than asserted. All five v1.0 incidents, at F1 to F5. The bid-seeding bracket, converted from a generator into a test. The doctrine lines that carry the reasoning: rank memory holds better on a taper than on a hard stop; the drift is the cost of waiting; UNKNOWN is a legal finding; PPC is neither credited nor blamed until the attribution check completes; and the diagnosis is scored separately from the prediction. Registered in the Supersession Map at publication.

10.3 Amendment Register: fifteen items for v4.1

Each item is a numeral or a rule identity the procedures above require and that no attached document supplies. None is written into this SOP as a threshold.

ID	The gap	Where it bites	What is needed, and the evidence available now
AM-1	No recovery velocity reference exists.	P7 step 3; every worked example.	Highest-priority item in this register. Family F6 bands velocity only in positions per week and supplies no per-class figure; the recovery duration calculation runs in positions per day. v1.0 asserted 2.0 per day attributed to v4.0 and it is not there. The figure is what converts the D10 classification into money: WE-1 shows a $1,592.96 difference in planned spend on one term between the recovery and cold assumptions. Needed: a reference by recovery class, or an instruction to treat duration as a checkpoint output rather than an input.
AM-2	No observation window per cause class.	P5 step 4; Section 2.	v1.0 instructed observation "per the v4.0 observation window for the cause class"; no such table exists. Family F13 carries persistence classes at syntax grain and rule T5 carries daily checks for 3 weeks on the post-restock mode only. Needed: a window per class, or an instruction that rank velocity turning positive closes the window regardless of elapsed time.
AM-3	No interval between the restoration artifact and the start of observation.	P4 step 7; Section 2.	Restorations propagate at different speeds: a restock is immediate, a price change reprices instantly, a listing edit re-indexes on the platform's schedule. Observing all three from the artifact date measures different things. Needed: either a per-class interval or an explicit instruction that the artifact date starts the clock in every class.
AM-4	Two exit tests resolve at different ranks with no precedence stated.	P13 step 1; WE-1.	Rule T5 exits the recovery mode at "baseline reached"; rules S-R6 and T1 fire the taper at 2 consecutive weeks at or under target rank. Where the pre-OOS baseline is better than the standing target these are different ranks on the same term, and WE-1 separates them by one position and fourteen days. Needed: a precedence rule, or an instruction that the recovery target is the baseline and the standing target is superseded for the duration.

ID	The gap	Where it bites	What is needed, and the evidence available now
AM-5	No re-dating rule when gate G10 suppresses the read a checkpoint requires, and no ceiling extension for the suppressed interval.	P3 step 7; P13; WE-1; WE-3.	Rules S-I5 and S-R13 set a 3-week checkpoint; gate G10 can make the read at that checkpoint inadmissible. WE-3 computes one instance at $893.08 of spend and $288.28 of loss across a 6-day extension that no rule sizes. The same shape appears on the S-R6 stability window in WE-1, where rule M2 requires the loss ceiling to be capped and dated and the window runs 14 days past its date. Needed: a re-dating rule and a statement of whether the extension is sized into the worksheet.
AM-6	No recovery count and no window for the serial-recovery escalation.	P14 step 7; F4; WE-4.	v1.0 asserted "the third recovery inside the v4.0 window"; v4.0 carries neither. WE-4 shows the pattern is invisible to individual post-mortems because each recovery scores HIT. Needed: a count and a rolling window, or an instruction that any second recovery on one term inside the review period triggers the read.
AM-7	The bid-seeding bracket constrains without determining.	P7 step 4; WE-1.	The discovered floor and the prior push CPC bracket the re-entry bid, and v1.0 wrote a value inside the bracket matching neither endpoint nor the midpoint. On the WE-1 sizing the bracket spans $0.55, which is $34.63 per day and $415.56 across a 12-day recovery. This SOP treats the bracket as a test on the M2-supplied Expected CPC. Needed: ratification of that method, or a rule for choosing inside the bracket.
AM-8	No maximum lag from funding release to standup outside the restock case.	Section 2; P11.	Family F3 ties the post-restock play to restock confirmation, which governs that lane. For the other six cause classes the release carries a date and nothing states how long the standup may trail it. A release that ages loses the queue position it won.
AM-9	No re-check interval after an indexing or listing restoration.	P4 step 6.	Gate G5 is binary and states no interval at which a re-check confirms the fix propagated. Recoveries have been sized against an indexing fix that had not yet taken effect. Needed: an interval, or a required artifact from the listing owner confirming the live state.
AM-10	No depth, rung count or interval for the pre-OOS taper.	P9 step 4.	Rule S-I3 states "taper to defense on the variation's terms" and sets no rung count, no interval and no floor for the pre-OOS case. #28 owns rung mechanics but the destination is not a numeral. Needed: a defense-floor definition for the pre-OOS lane, or an instruction that the standing defense floor governs.

ID	The gap	Where it bites	What is needed, and the evidence available now
AM-11	No buyability gate.	P10 step 3; P11 step 3.	Gate G4 covers eligibility and suppression. An offer can read eligible while the buy box is lost or the variation is unorderable, and spend against it buys nothing at all, which is the F5 emergency class. Needed: a buy-box and orderability condition inside G4 or as its own gate.
AM-12	No interval at which a flat recovery becomes a diagnosis miss.	P12 step 5.	v1.0 asserted day-4 and it traces to no source. Rule T5's daily checks and the 3-week checkpoint bound the read but state no point at which absence of movement re-opens P1 rather than continuing. Needed: an interval, or an instruction that only the 3-week checkpoint tests it.
AM-13	The recovery-zone test is defined two different ways.	P6 step 2.	Registered fresh, and distinct from Ledger conflict C17. C17 covers the identifier divergence (S-R12 against S-R13). This is definitional: the KDR Template v3.0 "Bands & Verdicts Reference" sheet defines the recovery zone as "within 10 of Best Rank"; v4.0 family F6 and rule D10 define it as best rank at or under 10 in absolute position. On a term whose best rank was 24 the two tests disagree completely, and the classification decides both funding priority and sizing. Needed: reconciliation in the same release that resolves C17.
AM-14	No cool-down before a downgraded term re-enters the queue.	P13 step 7.	Rule S-R13 downgrades a failed recovery to NEVER-RANKED realism; nothing prevents the same term returning the following cycle with the same assumptions and no changed evidence. Needed: an interval, or an explicit requirement that re-entry cite what changed.
AM-15	Record retention.	Section 8.	No v4.0 rule sets a retention period for the pre-OOS baseline, the cause record or the decision log, and the baseline in particular has an operational shelf life because rule D10 reads a trailing 12 months. Duplicates items logged by prior build sessions; v4.1 folds once.

11. Operator Ramp

Built from the PPC Authority Matrix v2.0 rows that name this component's decisions. Each row becomes a competence the incoming operator demonstrates, and the checkpoint holder named in the matrix reviews in week 2 and audits in week 4. Note the matrix vacancy rule: an unfilled role's decision right reverts to the PPC Lead and its execution to the PPC Manager, so a ramp in progress never leaves a cause confirmation unowned.

Authority Matrix row	Rights on that row	Week 1: observe	Week 2: supervised	Week 4: unsupervised, audited
Weekly verdicts (established US)	PPC Lead decides; incoming operator and PPC Manager execute; brand-management owner reviews.	Date three historical losses from the daily series without assistance and state each inflection with its interval. Match the checkpoint holder's independent dating on all three.	Run P1 and P2 on every loss in assigned scope. Name one primary cause with evidence and defend the classes ruled out, not only the one named.	Issue weekly recovery verdicts on assigned scope. The daily huddle review by the brand-management owner is review, not veto; disagreements route up rather than sideways.
Push funding (the M5 queue)	CEO decides above the envelope; PPC Lead decides within it; incoming operator executes.	Sit the funding call on every recovery candidate this cycle and state, for each, which of the two decision rights applies and why.	Compute cost-to-close and the M5 inputs for every candidate; the PPC Lead reviews the arithmetic before the call. Draft one worksheet with all three P7 adjustments visible and separated.	Own P6 through P8 end to end on assigned scope. The PPC Lead audits the ordering and the concessions rather than reproducing them.
Structure creation (all SOPs)	PPC Lead decides or per SOP scope; incoming operator executes.	Watch one post-restock standup end to end, including the buyability check, the reverse re-point, the wall session and the T+1 return.	Issue the standup order at P11 with the five carried values; the checkpoint holder confirms before it reaches SOP-12.	Issue standups on assigned scope. Every creation on a funded recovery remains human-mandatory at deployment.
Bulk deployments (#38)	PPC Lead reviews; incoming operator and PPC Manager execute.	Read one full staged set from a recovery and its T+1 diff. Identify which reason strings carry a scenario ID and which would fail staging.	Stage the change sets this SOP orders; the verification reconciliation is reviewed line by line.	Stage and reconcile unsupervised. A landed-value mismatch on a same-day restock deployment is routed the same day rather than absorbed.
Sibling declarations (S-series)	PPC Lead decides; incoming operator executes; brand-management owner reviews.	Read the family board and the open ownership question on at least one family root. State what gate G11 is currently blocking on a recovering term.	Supply both siblings' recovery class and cost-to-close for a family ordering; the PPC Lead makes the call.	Detect and hold sibling conflicts at P8 step 4 without prompting, and supply the ordering inputs unasked.
Event plans (#32)	PPC Lead decides with the deals calendar owner; operator executes.	Read one armed window's plan and compute its gate G10 guard end date. Identify which live recoveries have a checkpoint inside the guard.	Run P3 on a live recovery through a window under review: return the void with its class and hand it to #6 and #35 correctly.	Operate live recoveries through a full window and settle, unsupervised, and correctly refuse to open a recovery on a post-deal dip inside settle.
Code-red declaration and close	PPC Lead decides; incoming operator and PPC Manager execute.	Read the product posture rules and state, for each live recovery, what a push freeze and what Recovery Mode would each do to it.	Apply the rule R3 exception correctly on a product in Recovery Mode: identify which recovery candidates qualify and which do not, and why both conditions matter.	Apply the tightened Recovery Mode confirmation thresholds without being reminded that they replaced the standing ones.

Authority Matrix row	Rights on that row	Week 1: observe	Week 2: supervised	Week 4: unsupervised, audited
Terminal and LTSF PPC lane	PPC Lead decides with the LTSF owner; operator executes; brand-management owner reviews.	Identify which recovering terms sit on products carrying a CONSTRAINED earning-potential class and state what rule S-A10 does to them.	Route one CONSTRAINED case correctly: stop at the boundary, state the dependency, and do not infer the tier rules that sit outside this corpus.	Route by class without error. Funding a recovery on a CONSTRAINED product is a governance failure even where the arithmetic is correct.

11.1 Live-defense questions

These test the reasoning behind the thresholds rather than recall of their values. The operator answers cold, with the workbook open and the criteria document closed. An operator who can recite "best rank at or under 10" and cannot say why the classification changes the sizing has not passed.

1. A head term fell from rank 6 to 22 and the product log shows a stockout in the same three-week period. State the one comparison that decides whether the stockout caused it, state which direction admits the class and which excludes it, and say what you do when the rank series for that term is weekly rather than daily.
2. Two terms both lost rank, and neither product log window contains an event. Explain why that finding is evidence rather than an absence of evidence, name the two cause classes it points at, and name the log you would open next that is not T5.
3. A recovery reached its target rank on day 12 and its pre-OOS baseline is one position better. State which exit test applies, what the other one says, and why you write both down.
4. Your worksheet assumed 2.0 positions per day and the recovery realized 1.9. CVR and cost both came in within 3% of plan. Say what you change for the next recovery on this term class and what you would have to be careful not to change.
5. An event window opens on day 9 of a ten-day recovery whose checkpoint is day 21. Compute the guard end, say what the day-21 read is worth, and say what you file the score as and why that is not the same as saying the recovery failed.
6. Explain why a recovery push on top of a rank that is already climbing is a worse mistake than a recovery push that is slightly undersized.
7. A term lost four positions and the loss dates to one of our own ladder steps. State the remedy, state what the remedy costs against the alternative in dollars per week, and say what happens to the floor register.
8. The price cause class cannot be tested because a feed is missing. Explain in one sentence why you write "not testable" rather than "not found", and say what breaks downstream if you write the second.
9. A product is in Recovery Mode and every new Ranking objective is frozen. Explain why your recovery candidate may still be funded, and name both conditions it has to meet.
10. Name three things this SOP is forbidden to decide and say which component decides each.

12. Interface Contracts

Eighteen handoffs, split upstream and downstream. Every row states what happens when the handoff does not arrive, which is the clause the corpus has always been missing. Sources: the Cross-Reference Ledger row for component 24 and the rows of every counterpart named, cross-checked against Master PPC Data Architecture v1.0 for artifact names and tab locations. Where a counterpart was rebuilt in the same wave as this document, the seam is written from its Ledger row rather than from a procedure number, because those numbers move.

ID	Counterpart	Artifact and timing	Non-arrival consequence
I-1	SOP-23, upstream	The pre-OOS baseline snapshot, the D10 recovery class, the proven ceiling, and the floor and prior-push CPC history. At candidacy hand-over and at any push close.	The highest-consequence non-arrival on the upstream side. The term misclassifies as NEVER-RANKED and loses the funding priority its proven ceiling earned, and the recovery sizes cold: on the WE-1 distance that is $1,592.96 of planned spend the account did not need to commit. There is no local workaround, because the baseline can only be written before the outage and cannot be reconstructed afterwards.

ID	Counterpart	Artifact and timing	Non-arrival consequence
I-2	Inventory feed and Data Operations, upstream	Cover, adequacy bands, projected days to stockout and restock arrival dates into T2, daily per the Economics and Context Pack cadence.	The one precondition with no workaround. Without it the P9 trigger never fires, so no baseline is written and the whole recovery lane degrades to reactive. Restock dates that drift without updating leave a standup planned against a stale date, which is how P11's day-0 discipline fails without anyone deciding to abandon it. Per Ledger conflict C3 this interface names the feed and the instrument, never a #31 procedure.
I-3	#39 and Data Operations, upstream	T7 manifest statuses for A1, A3, A6, B3, D1, D2, D3, D6, D8 and D9, weekly, with the daily rank series refreshing daily.	P1 cannot date a loss outside D9's head-term coverage and P2 cannot test the price or competitor classes at all. The failure is quiet and specific: the diagnosis returns UNKNOWN on terms whose cause is in fact readable from data the account does not hold, so the register of unexplained losses grows and reads as a diagnostic weakness rather than a feed gap. WE-2 attaches $1,256.67 of held recovery to the D3 and D8 rows for this reason.
I-4	Brand Management, upstream	The T5 product log at completeness: listing, image and A+ changes, price moves, out-of-stock windows per variation, deal windows, competitor entries, targeting changes, eligibility flips and fee-dimension changes, entered dated.	P1 and P2 both stop. This SOP owns reading the log and Brand Management owns filling it, and an unlogged event is indistinguishable from no event, which converts a knowable cause into UNKNOWN. Every one of the seven classes except taper too deep depends on this artifact. A log with gaps is more dangerous than an empty one, because the empty-log finding at P1 step 7 is only valid if the log is trusted to be complete.
I-5	#28, upstream	The reversion rule and rung values for the pre-OOS taper, the floor register with its dated entries and competitive context notes, and exit-bid references. On request per overlay and per restore.	The P9 taper cannot stage and the taper-too-deep class has no remedy to issue, so a loss whose correct fix costs $6.60 a week routes to a sized push at $392.00. The floor register is also the instrument P1 reads to detect the class at all: without dated rungs, a ladder-caused loss and a market-caused loss are indistinguishable. Dependency D3 is OPEN, and #28's Ledger row names this SOP among the taper deciders while carrying no reciprocal interface row for it, which is reported at Section 13.
I-6	#32, upstream	The armed window with its dates, the settle marks, and the pre-window baseline snapshot on any product carrying a live or candidate recovery. At arming and at settle.	P3 cannot compute the guard end, so voids are not detected and confounded reads file as MISS. Two consequences compound: this SOP's own scores corrupt, and #6 and #35 both receive contaminated inputs they cannot detect. Post-deal dips inside settle also open recoveries they should not, funding a calendar effect at push economics.

ID	Counterpart	Artifact and timing	Non-arrival consequence
I-7	#41 economics, upstream	The per-variation margin table with its freshness state, repointed into T0 zone C. Monthly plus event-driven refreshes, before the Monday packet-ready read.	Every ceiling in this SOP carries PROVISIONAL and every worked example runs on the training base. This is external dependency D1, open, and the Ledger names it as capping this component. Recoveries still size, but no ceiling is defensible on live economics and the margin-dollar class that gates the O3 THIN clause is blended rather than per-variation.
I-8	#33, upstream	The family term-ownership declaration on any shared recovering term, with the lead product and the support or stand-down posture. On conflict detection.	Gate G11 holds both restore steps indefinitely. Neither sibling recovers and the term stays lost, which is worse than either sibling recovering alone. The Bamboo root ownership was left OPEN at cycle 1, so this is a live condition rather than a hypothetical.
I-9	#37, upstream	The rule change log covering any loss window under diagnosis, on request at P2 step 6.	The automated half of the taper-too-deep class is invisible. A runaway rule stepping a bid against a stale target produces exactly the ladder-date signature P2 looks for in T6, without writing a T6 row, so the loss reads as an unexplained market event and the rule keeps running through the recovery it caused. SOP-26 records the pattern: one rule stepped a defensive term six rungs and nothing alarmed because each step was inside its own limit.
I-10	Listing, pricing and review owners, upstream	The restoration artifact per cause class: the live page state after a listing revert, the restored price with its Price Position Index recomputed, the eligibility flag, or the review-velocity plan.	P4 cannot close and no recovery funds, which is the correct failure and the expensive one. A recovery held on an unreturned listing artifact is a rank not recovered for a reason that has nothing to do with PPC, and the days are quantified in T6 against the owner rather than absorbed silently by the PPC queue.
I-11	SOP-26, upstream	The product Recovery Mode state and the rule R3 override set, at declaration and at exit.	P8 step 2 cannot apply the R3 exception, so either every recovery is frozen on a code-red product, which removes the one funding path the rule deliberately preserves, or recoveries fund without the HIGH-confidence condition the exception requires. Both are governance failures and they point in opposite directions.
I-12	SOP-25, upstream and downstream	Outbound: the cause classes and the recovery gate. Inbound: the descent schedule and its dated rungs. Continuously, and per the Ledger row for #25.	Its failure mode F2 exists precisely because a descent slip and a recovery get confused. Without the cause classes flowing outbound, a scheduled descent rung reads as a loss and opens a recovery against our own plan; without the descent schedule flowing inbound, this SOP cannot rule the taper-too-deep class in or out. The Ledger records the recovery gate on its side as an interface whose absence is the gate working, which only holds while the classes are published.

ID	Counterpart	Artifact and timing	Non-arrival consequence
I-13	#6, downstream	The confound cause classes, the void determinations with their class references, and the diagnosis score at every close.	SOP-06 defers its confound classes to this procedure by name, so without them no VOID is legal and every confounded read files as a MISS. Rule-level accuracy under rule V5 then computes against contaminated scores, and a rule that is working correctly enters review for failures the world caused. The VOID abuse watch at #6's F3 also loses its reference set.
I-14	#35, downstream	The void classes, and any confound references where an incrementality read shared a voided window.	A confounded incrementality read is passed over as clean and the continue-or-restore call is made against it, which sets a defended floor on evidence that was never valid. A confounded read handed over as clean is worse than no read.
I-15	SOP-12, downstream	The standup order carrying campaign, re-entry bid, daily budget, objective tag and checkpoint date; the re-point instruction; the ladder-restore instruction; the retag at close. Same day on a restock; same cycle otherwise.	A creation with no verdict-carried bid returns as incomplete and the restock day passes unspent, which is the F2 drift condition at $95.20 per day on the WE-1 sizing. On the retag side, non-arrival is F11: the recovered term reads as a live push to every rollup that counts objectives.
I-16	#28, downstream	The recovery's achievement date and bid at the taper handoff; the restored floor after a taper-too-deep correction; and the failed-probe marking that keeps a slipped rung out of the register.	The ladder starts from an unstated bid and the first rung is guessed. Worse on the correction side: a probe that failed silently lowers the recorded floor, so the register carries a price the position does not hold and every future drift check measures against it. The error compounds quarterly.
I-17	Instrument layer, downstream	The queue submission with cost-to-close into T0 zone H and Account Rollup tab A2; the released daily budget line on any concession, downgrade or taper. Same cycle as the verdict.	Freed budget stays notionally allocated to a recovery that has ended, so the queue under-reports available dollars and the next candidate is declined for budget that exists. Per Ledger conflict C3 this names the instrument and the tab; there is no #31 procedure to escalate against and a stalled release routes to the PPC Lead by role.
I-18	#38 and the verification layer, downstream	Staged change sets carrying verdict and scenario identifier in every reason string, with the T+1 diff returning. Tuesday by default; same day on a restock standup or a RED flip.	A staged row missing its scenario identifier does not deploy, so the change silently does not happen while the T6 row claims it did. This is acute on the same-day lanes: a re-point that did not land leaves spend pointed at an unbuyable variation, which is the F5 emergency class arriving through a deployment gap rather than through a decision.

13. Reported Findings Beyond the Pre-Declared List

Anti-drift rule 5 requires a build session to report conflicts it finds beyond those already registered. Conflict identifiers are assigned at close-out from the current register and never inside a session, so these are reported as findings without IDs.

1. The recovery-zone test is defined two different ways, which is distinct from C17. C17 records the identifier divergence between S-R12 and S-R13. This is a definitional divergence in the same neighbourhood: the KDR "Bands & Verdicts Reference" sheet defines the recovery zone as "within 10 of Best Rank" while v4.0 family F6 and rule D10 define it as best rank at or under 10 in absolute position. The two tests select different term sets. Logged here as AM-13 and reported for registration.

2. #28 names this component among the taper deciders and carries no interface row for it. The Ledger row for component 28 states that whether a taper happens and when belongs to #12 P4, #23, #24, #25 and #26, and its 22 interface contracts carry rows for #23, #25 and #26 and none for #24. This SOP writes the seam from its own side at I-5 and I-16, and the reciprocal rows are owed at #28's next increment.
3. SOP-12 v2.0 Section 5 maps RECOVERY PUSH to S-R12, which is the C17 inversion appearing in the exemplar itself. Reported not as a new conflict but as a located instance, since C17's resolution names SOP-12 v2.0 Section 5 as one of the surfaces to correct in v4.1 and this session confirms it is still present.
4. The Data Architecture Section 6.1 carries the same inversion in the opposite direction, mapping the zero-exact structure filter to S-R13 where v4.0 assigns it to S-R12. Within C17's scope; reported so the correction list is complete.

Pack-state note. The version note attached to this session records the Ledger at v2.5 with Wave 5A open on four rows. The Ledger reads v2.8: sessions 17, 18 and 19 have closed and this component is the only Wave 5A row outstanding. The Ledger states that its row closes the wave at v2.9, which is also the Wave 5B gate, and the row returned with this document carries that version accordingly.

Companions: Master PPC Decision Criteria System v4.0 (Section 3 families F1, F3, F4, F5, F6, F13, F15, F16; Section 4 rules X1, X4, X5, X8, X9, X10; Section 5 gates G1-G12; Section 6 rules O3, O5, O7; Section 7 rules D5, D6, D7, D9, D10; Section 8.1 S-A3, S-A4, S-A10; Section 8.3 S-I2 to S-I6; Section 8.4 S-R1 to S-R16; Section 8.12 S-E5; Section 9 levers L1, L2, L5, L9; Section 10 M2, M3, M5, M10; Section 11 T1, T5, T8, R1-R6; Section 12 P1, P5, P8; Section 13.1 the closed verdict vocabulary; Section 13.3 rule J7; Section 14 V1, V4, V5; Section 16 Worked Pass 3) · Keyword Decision Record Template v3.0 "Bands & Verdicts Reference" (the quoted verdict strings and the campaign subset) · PPC Strategic Doctrine v1.0 principles 3 and 4, Section 6.3 and Section 14 (the why) · PPC Economics Worked-Example Set v1.0 cases E3 and E5 (the WE-1 and WE-4 anchors) · PPC Authority Matrix v2.0 Section 1 (Elements 9 and 11) · Master PPC Data Architecture v1.0 Sections 4, 6, 9, 11 and 12 and the Command Workbook (artifacts and tabs) · SOP-23 (parent), SOP-12 (execution), SOP-25 (post-objective), SOP-26 (Recovery Mode posture), SOP-06 (scoring), #28 (ladder), #29 (walls), #32 (windows), #33 (family ownership), #35 (incrementality), #37 (rules), #38 (staging), #39 (feeds), #41 (economics) · PPC Cross-Reference Ledger v2.8 (component 24 row; conflicts C1 to C23).